



Revisiting the observability and identifiability properties of a popular HIV model

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ABSTRACT

This paper revisits the observability and identifiability properties of a popular ODE model commonly adopted to characterize the HIV dynamics in HIV-infected patients with antiretroviral treatment. These properties are determined by using the general analytical solution of the unknown input observability problem, introduced very recently in Martinelli (2022). This solution provides the systematic procedures able to determine the *state observability* and the *parameter identifiability* of any ODE model, in particular, even in the presence of time varying parameters. Four variants of the HIV model are investigated. They differ because some of their parameters are considered constant or time varying. Fundamental new properties, which also highlight an error in the scientific literature, are automatically determined and discussed. Additionally, for each variant, the paper provides a quantitative answer to the following practical question: *What is the minimal external information (external to the available measurements of the system outputs) required to make observable the state and identifiable all the model parameters?* The answer to this fundamental question is obtained by exploiting the concept of continuous symmetry, recently introduced in Martinelli (2019). This concept allows us to determine a first preliminary general result which is then applied to the HIV model. Finally, for each variant, the paper concludes by providing a redefinition of the state and of the parameters in order to obtain a full description of the system only in terms of a state which is observable and a set of parameters which are identifiable (both constant and time varying).

1. Introduction

Ordinary differential equations are an essential tool for investigating many biological phenomena (DiStefano, 2015). They are commonly used to describe the time evolution of many biological systems. An ODE model is often characterized by several parameters, which can be known or unknown and can be constant or time varying. State observability and parameter identifiability are two structural properties of an ODE model. The former characterizes the possibility of inferring the values that the state components take, starting from the knowledge of the system inputs and outputs (e.g., Kalman, 1961, 1963; Hermann and Krener, 1977; Casti, 1982; Isidori, 1995). The latter characterizes the possibility of inferring the values of the unknown model parameters, again, from the knowledge of the system inputs and outputs (e.g., Grewal and Glover, 1976; Tunali and Tarn, 1987; Walter and Pronzato, 1996). Performing the observability and identifiability analysis is fundamental in order to set up an ODE model, i.e., to correctly define the state and the parameters that the information contained in the sensor measurements allows us to estimate. Characterizing a given system with a state that is unobservable and/or with parameters

that are unidentifiable can have dramatic consequences during the estimation and/or the identification process.

In this paper we revisit the observability and identifiability properties of a popular ODE model that characterizes the HIV dynamics in HIV-infected patients with antiretroviral treatment. This model was introduced during the 90's by Perelson and collaborators (e.g., Perelson and Kirschner, 1993; Ho et al., 1995; Perelson et al., 1997; Perelson and Nelson, 1999; Perelson, 2002). The model is characterized by the following three dynamics equations:

$$\begin{cases} \dot{T}_U &= \lambda - \rho T_U - \eta T_U V \\ \dot{T}_I &= \eta T_U V - \delta T_I \\ \dot{V} &= N \delta T_I - c V \end{cases} \quad (1)$$

where, T_U is the concentration of uninfected cells, T_I the concentration of infected cells, and V the viral load. The model is also characterized by the six parameters λ , ρ , η , δ , N , c . They are defined as follows:

- λ is the source rate of uninfected cells.
- ρ is the death rate of uninfected cells.

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Table 1

The four variants studied in this paper (TV stands for time varying).

Variant	Const. parameters	TV parameters	Outputs
1	$\lambda, \rho, \delta, c, N$	η	$V, T_U + T_I$
2	$\lambda, \rho, \delta, c, \eta$	N	$V, T_U + T_I$
3	λ, ρ, δ, c	η, N	V, T_U, T_I
4	λ, ρ, δ, c	η, N	$V, T_U + T_I$

- η is the infection rate.
- δ is the death rate of infected cells.
- N is the average number of virions produced by a single infected cell during its lifetime.
- c is the clearance rate of free virions.

In this paper, we assume that all these parameters are unknown. Previous observability and identifiability analyses studied the case when all the parameters are constant and the case when the parameter η is time varying. When the system outputs are $y_1 = V$ and $y_2 = T_U + T_I$, the results available in the scientific literature establish that, in both cases, all the parameters are identifiable and the state is observable (e.g., see Miao et al., 2011). This is actually erroneous. Specifically, while it is true that in the first case all the parameters are identifiable and the state is observable, in the second case, the time varying parameter $\eta(t)$ and the two constant parameters δ , and N , are unidentifiable. In addition, also the two states T_U and T_I are unobservable when η is time varying. This error has had some damaging consequences. In particular, several authors have spent a great deal of effort to perform the impossible identification of the unidentifiable parameters. For instance, in Hartmann et al. (2012), the authors introduced methods based on the Extended Kalman Filter and on the Particle Filter to estimate the state and the parameters of the model in both the aforementioned cases. Not surprisingly, they obtained good performance only in the first case (i.e., when all the parameters are constant and the model is really observable and identifiable), while, in the second case (time varying η) the performance deteriorates. Note that, in other works that assumed η time varying and that introduced successful methods for its estimation (e.g., in Chen and Wu, 2008), the outputs consisted of all the state components, i.e., T_U , T_I , and V , instead of only $y_1 = V$ and $y_2 = T_U + T_I$ (with the three outputs all the model parameters are identifiable, as we prove in Section 3.4).

In this paper, we apply the procedures summarized in Section 2 (Algorithm 1 and Algorithm 4) to study the observability and identifiability properties of several variants of the model characterized by the dynamics in (1). In particular, based on Mhaweje et al. (2010), where the parameter N was also considered time varying, we define the four variants in Table 1.¹ Sections 3, 4, 5, and 6, provide an exhaustive study of the observability and identifiability properties of all these variants, by using Algorithms 1 and 4. Additionally, in the presence of unobservability and/or unidentifiability, we provide some numerical results by generating values of the unobservable components of the state and values of the unidentifiable parameters which are indistinguishable from their true values. We also directly show that these indistinguishable values reproduce exactly the same outputs. This is an independent and unequivocal proof of the validity of all the observability and identifiability properties determined in Sections 3–6. For each variant, we also provide the minimal information that must be added to the knowledge of the outputs, in order to make the

state observable and all the parameters identifiable. Finally, for each variant, we conclude by providing a full description of the system only in terms of a state which is observable and a set of parameters which are identifiable.

In addition to these new findings about the ODE model in (1), this paper provides a complete implementation of Algorithms 1 and 4 and presents a new general approach to observability/identifiability that allows us to characterize the minimal information required to make observable and/or identifiable a model in the presence of unobservability and/or unidentifiability (Section 2.9). Algorithms 1 and 4 automatically determine the observability and identifiability properties of any ODE model, independently of its complexity and type of nonlinearity. We wish to emphasize that the model may include the presence of parameters that vary with time. From a mathematical point of view, an unknown time varying parameter plays exactly the same role of a system input that is unknown. Not surprisingly, Algorithms 1 and 4 are profoundly based on the very recent solution of the unknown input observability problem, which has remained an open problem in the last half century. In this paper we use the term time varying parameter and unknown input (or also UI) interchangeably.

The first important analytical condition to check the state observability was introduced in the seventies and it is called the *observability rank condition* (Hermann and Krener, 1977; Casti, 1982; Isidori, 1995). The observability rank condition is a fundamental result that has extensively been used in many application domains, ranging from computer vision (e.g., Paul and Roumeliotis, 2018), robotics (e.g., Hesch et al., 2014; Schiano and Tron, 2018; Yang and Huang, 2019), calibration (e.g., Guo and Roumeliotis), mechanical engineering (e.g., Chatzis et al., 2015) up to biology (e.g., DiStefano, 2015; Miao et al., 2011; Villaverde, 2019; Villaverde and Barreiro, 2016; Chis et al., 2011) and chemistry (e.g., Elias August 1 and Papachristodoulou, 2009). The observability rank condition can deal with any system, independently of its complexity and type of nonlinearity. It works automatically, i.e., it can be verified by following the steps of a systematic procedure that does not require supervision (it only requires to perform differentiation and to compute matrix ranks). On the other hand, the observability rank condition presents a limitation: it does not account for the presence of unknown inputs (or time varying parameters). This is a non-negligible limitation. The problem of unknown input observability (i.e., the problem of obtaining the analytical and automatic criterion that extends the observability rank condition to the case when the model includes time varying parameters) was defined long time ago (Basile and Marro, 1969; Guidorzi and Marro, 1971) and remained unsolved for half a century. The control theory community has primarily focused the attention on the design of observers for both linear and nonlinear systems in the presence of unknown inputs, in many cases in the context of fault diagnosis. In this framework, interesting conditions for the existence of an unknown input observer were introduced. On the other hand, these conditions have the following restrictions/inconveniences:

1. They refer to a restricted class of systems. In particular, the considered systems are often characterized by linearity (or some specific type of nonlinearity) with respect to the state in some of the functions that characterize the dynamics and/or the system outputs.² None of these conditions refers to any type of nonlinearity in the aforementioned functions.
2. Their application is laborious, as they must be adapted to the specific case and this adjustment requires inventiveness by the user. In particular, this process cannot be carried out by simply following the steps of a systematic procedure (e.g., by running a code without supervision).

Identifiability analysis is also present in a large variety of scientific domains and several interesting methods have been proposed. An

¹ Note that we already provided the study of Variant 1 in Martinelli (2023b). However, the analysis in Martinelli (2023b) was only limited to the identifiability properties of this variant and only provided the results of the observability analysis. Here, we also include all the steps of the observability analysis. In addition, we adopt a different data set to provide some numerical results. Finally, here we also provide a full description of the system only in terms of an observable state.

² These are the functions that appear in (2) i.e., the functions f and h .

exhaustive review of the methods proposed up to 2011 can be found in Miao et al. (2011). Some of these methods are interesting and, in many cases, they can successfully be used to detect all the identifiability properties of a given ODE model. However, they present the same two fundamental aforementioned limitations. In particular, regarding the first limitation, note that many of them can be only used when the nonlinearity is polynomial. Regarding the second limitation, note that it becomes particularly relevant in the presence of time varying parameters where, in some cases, an erroneous application of these methods has produced wrong results. Specifically, this was the case of the aforementioned erroneous identifiability analysis of the HIV model in (1) with η time varying, carried out in Section 6.2 of Miao et al. (2011), which was obtained by using one of these methods (a detailed explanation of the error made by the authors of Miao et al. (2011) is available in Martinelli (2023a). In Section 3, we provide a thorough observability and identifiability analysis for this model and we determine the true observability and identifiability properties.

In the last decade, there has been a significant progress in the Unknown Input Observability problem (UIO). Specifically, the nonlinear UIO problem was approached by introducing an extended state that includes the original state together with the unknown inputs and their time derivatives up to a given order (Belo et al., 2010; Martinelli, 2015; Villaverde et al., 2019; Maes et al., 2019; Villaverde et al., 2016). In Villaverde et al. (2019), Maes et al. (2019), Villaverde et al. (2016), several automatic iterative algorithms were introduced. Note that some of these algorithms were developed to study several biological systems since, very often, measuring some of the inputs of these systems can be challenging. These algorithms work automatically and are able to check both the state observability and the identifiability of the model parameters. On the other hand, they suffer from the following limitations:

- They do not converge, automatically. In particular, at each iterative step, the state is extended by including new time derivatives of the unknown inputs. Consequently, if the extended state is observable at a given step, convergence is achieved. However, if this were not the case, we can never exclude that, at a later step, the extended state becomes observable. Therefore, in the presence of unobservability, all these algorithms remain inconclusive.
- Due to the previous state augmentation, the computational burden can easily become prohibitive after a few steps.

Very recently, we introduced the analytical solution of the nonlinear UIO (Martinelli, 2022), starting from the results/derivations presented in Martinelli (2020).³ In particular, the solution introduced in Martinelli (2022) does not encounter the aforementioned limitations⁴ and provides a full answer to the problem of the state observability for any nonlinear ODE model, in the presence of unknown inputs (or time varying parameters). The solution introduced in Martinelli (2022) is the solution of the unknown input observability problem. In Martinelli (2022), we also exploited this analytical solution to provide an important analytical result that characterizes the parameter identifiability.⁵ (unknown input reconstruction).

This paper is structured as follows. Section 2 provides the compact presentation of the two procedures to automatically obtain the state observability (Algorithm 1) and the parameter identifiability (Algorithm

4). In addition to these two algorithms, Section 2 provides a compact summary of our recent results provided in Section 4.6 of Martinelli (2020). These results establish that state unobservability is always the consequence of a system invariance under one or more continuous transformations (called *continuous symmetries*). This summary is provided in Section 2.4, where it is also described how to algebraically determine all the continuous symmetries starting from the observability codistribution returned by Algorithm 1. Furthermore, in Section 2.4 we provide the analytical method to determine infinitely many indistinguishable states in the presence of unobservability (this method consists of solving the differential equations in (15)). In Sections 2.6 and 2.7 we extend this analysis to the unknown inputs and we provide a fundamental theoretical result which is the analytical method to generate infinitely many indistinguishable values of the time varying parameters which are unidentifiable (in this case, the method consists of solving the differential equations in (30)). The concept of continuous symmetry is also exploited in Section 2.9 to determine a preliminary and partial result in order to characterize the missing information to make observable and/or identifiable a system in the presence of unobservability and/or unidentifiability. Finally, Section 2 introduces an elementary example, called the *case study* (Section 2.3). Throughout Section 2, the case study is extensively used to illustrate all the theoretical concepts summarized in this section, i.e., the functioning of Algorithms 1 and 4 and the analytical generation of the indistinguishable states and time varying parameters in the presence of unobservability and/or unidentifiability. For the case study, the results obtained by applying all these concepts become very intuitive. Sections 3–6 investigate the observability and identifiability properties of all the variants listed in Table 1 of the popular HIV model also studied in Perelson and Nelson (1999), Perelson (2002), Miao et al. (2011), Chen and Wu (2008), Villaverde et al. (2019). This study is based on the theoretical results presented in Section 2. Section 7 provides a comparison of our approach (which is summarized in Section 2) with existing popular approaches. Special focus is on the discrepancies in the results obtained for Variant 1. Finally, Section 8 provides our conclusion.

2. Systematic procedures to obtain the state observability and the parameter identifiability

This section gives a concise summary of the two systematic procedures that automatically determine the state observability and the parameter identifiability of any ODE model. The interested reader may find all the theoretical foundation and proofs in Martinelli (2022). We start by characterizing the ODE model in Section 2.1. Then, we provide the two systematic procedures in Sections 2.2 and 2.5, respectively (Algorithm 1 and Algorithm 4). In Section 2.3 we introduce the aforementioned case study. In Section 2.4, we provide a summary of Section 4.6 in Martinelli (2020), where we characterized the concept of continuous symmetry and, in Sections 2.6, and 2.7, we extend this concept to the unknown inputs. Section 2.9 provides the aforementioned result to characterize the missing information that makes observable/identifiable a system in the presence of unobservability and/or unidentifiability.

2.1. System characterization

Algorithms 1 and 4 automatically determine the observability and identifiability properties of any nonlinear system. Indeed, the model is characterized by the following general equations:

$$\begin{cases} \dot{X} &= f(X(t), t, U(t), Q, W(t)) \\ y &= h(X(t), t, U(t), Q, W(t)), \end{cases} \quad (2)$$

³ This analytical solution is also based on the results obtained in Martinelli (2019) and Sarafrazi et al. (2022), which deal with special systems, namely characterized by a single unknown input, and dynamics linear with respect to this unknown input.

⁴ Note that the solution introduced in Martinelli (2022) could need the inclusion of some of the UIs in the state. However, this inclusion is targeted and terminates in a finite number of steps, as soon as the extended system achieves its *highest unknown input degree of reconstructability* (see Section 5 in Martinelli (2022)).

⁵ The complete result was given in Martinelli (2023b).

where: $X(t) \in \mathbb{R}^m$ is the state,⁶ $y(t) \in \mathbb{R}^p$ the output vector, $U(t) \in \mathbb{R}^{m_u}$ the known system input vector, $Q \in \mathbb{R}^q$ the set of the unknown constant parameters, and $W(t) \in \mathbb{R}^{m_w}$ the set of the unknown time varying parameters. Note that $W(t)$ can be regarded as a system input vector that, precisely as $U(t)$, acts on the system dynamics. However, it differs from $U(t)$ in two fundamental respects: (i) it cannot be assigned, and (ii) it is unknown.

To exploit the theoretical results introduced in Martinelli (2022), instead of (2) we adopt the following system characterization:

$$\begin{cases} \dot{x} &= g^0(x, t) + \sum_{k=1}^{m_u} f^k(x, t)u_k(t) + \sum_{j=1}^{m_w} g^j(x, t)w_j(t) \\ y &= [h_1(x, t), \dots, h_p(x, t)], \end{cases} \quad (3)$$

where:

- $x \in \mathbb{R}^n$ is the state.
- $y \in \mathbb{R}^p$ is the output vector.
- $u_1(t), \dots, u_{m_u}(t)$ are the known inputs.
- $w_1(t), \dots, w_{m_w}(t)$ are the unknown inputs or the unknown time varying parameters.
- $f^1, \dots, f^{m_u}, g^0, g^1, \dots, g^{m_w}$ are $m_u + m_w + 1$ vector fields, which are assumed to be smooth functions of x and t .

It is immediate to realize that (3) is a special case of (2). In particular, it is obtained by requiring that the dynamics are affine in the inputs (both known and unknown) and the output function only depends on the state (x) and on the time (t). On the other hand, it is possible to show that the two characterizations are actually equivalent. In particular, we can easily convert (2) to (3). This is obtained by setting $u(t) = \dot{U}$, $w(t) = \dot{W}$ (with $u = [u_1, \dots, u_{m_u}]$ and $w = [w_1, \dots, w_{m_w}]$) and by including $U(t)$, $W(t)$ and Q in the state (i.e., $x = [X^T, U^T, W^T, Q^T]^T$). In the rest of this paper, we directly refer to the characterization given in (3).

2.2. Procedure for state observability: Algorithm 1

The procedure consists of a recursive algorithm (Algorithm 1⁷) that, given a system characterized by (3), automatically builds the entire observation space in a finite number of iterations. In particular, it builds the so-called *observability codistribution*, which is, by definition, the span of the gradients of all the observable functions.⁸ Algorithm 1 builds the generators of the observability codistribution. To this regard, note that, in the algorithm, statements like “ $\Omega = \text{span} \{ \nabla h_1, \dots, \nabla h_p \}$ ” (e.g., Line 2 of Algorithm 1) require no action. The algorithm builds the codistribution by building a set of its generators. When it recursively updates the codistribution it simply computes its new generators (and this is obtained by applying certain operations directly on the generators of the previous codistribution or by computing new covectors starting from the quantities that define the original system Σ in (3)).

The reader is addressed to Martinelli (2022) to obtain all the proofs that validate Algorithm 1 together with all the ingredients necessary for its implementation (a quick presentation of all these ingredients is

⁶ In general, instead of $\mathbb{R}^m$, the state belongs to a differential manifold of dimension m . This also holds for the other quantities that appear in (2) (and also in (3)), with the appropriate dimension. In this paper, for the sake of simplicity, we avoid the concept of differential manifold (the reader familiar with this concept can simply replace the Euclidean space of a given dimension with the differential manifold of the same dimension).

⁷ Algorithm 1 uses the automatic procedures provided by Algorithm 2 and Algorithm 3. Often, when we refer to Algorithm 1, we actually mean all the three algorithms (1, 2, and 3).

⁸ An observable function is a scalar function of the state x (for time varying systems also of t) whose value, at a given time t_0 , can be expressed in terms of the values that the known inputs and the system outputs take on a given time interval that includes t_0 . The definition of observation space is provided in Martinelli (2019), where Definition 2 defines this concept starting from the concept of indistinguishability. An observable function is precisely an element of this function space.

available in Chapter 3 of Martinelli (2023c)). Here, we only provide a quick description of the functioning of Algorithm 1 and a description of its outputs.

Algorithm 1: The algorithm to obtain the state observability in the presence of time varying parameters (or unknown inputs). Comments are in blue.

```

1: Set  $\Sigma$  the system in (3).
2:  $\Omega = \text{span} \{ \nabla h_1, \dots, \nabla h_p \}$ .
3: boolean Continue=true.
4: while 1 { This is the MAIN loop } do
5:   if  $\deg_w(\Omega) == m_w$  then
6:     break main loop.
7:   end if
8:    $m = \deg_w(\Omega) - 1$ .
9:   while  $\deg_w(\Omega) > m$  &  $\deg_w(\Omega) \neq m_w$  do
10:     $m = \deg_w(\Omega)$ .
11:     $[\tilde{h}_1, \dots, \tilde{h}_m] = S(\Sigma, \Omega)$ ,  $\Sigma = \mathcal{R}(\Sigma, \tilde{h}_1, \dots, \tilde{h}_m)$ .
12:    Compute  $\mu, \nu, \hat{g}^\alpha$  (Eqs (4)-(5)).
13:     $\Omega = \left\langle f^1, \dots, f^{m_u}, \left\{ \begin{matrix} \hat{g}^0, & \dots, & \hat{g}^m \\ g^{m+1}, & \dots, & g^{m_w} \end{matrix} \right\} \middle| \Omega \right\rangle$ 
14:   end while
15:   if  $\deg_w(\Omega) == m_w$  then
16:     break main loop.
17:   end if
18:   if  $m_u \neq 0$  then
19:     Run Algorithms 2 and 3 to compute  $\hat{k}^m = s_x^m + r^m$ .
20:   else
21:      $\hat{k}^m = 0$ .
22:   end if
23:   Set  $k = 0$  and  $\Omega_0 = \Omega$ 
24:   while 1 { This is the NESTED loop } do
25:      $k = k + 1$ .
26:     Reset  $[\Sigma, \Omega_{k-1}] = \mathcal{A}(\Sigma, m, \Omega_{k-1})$ .
27:     Compute  $\mu, \nu, \hat{g}^\alpha$  (Eqs (4)-(5)).
28:      $\Omega^* = \Omega_{k-1}$ .
29:     if  $(k \leq \hat{k}^m) \& (k \geq 2)$  then
30:        $\Omega^* = \Omega^* + \sum_{q=1}^m \sum_{i=1}^{m_u} \sum_{\alpha_1=0}^m \dots \sum_{\alpha_{k-1}=0}^m \text{span} \{ \mathcal{L}_{m[f^i]}^{(\alpha_1, \dots, \alpha_{k-1})} \nabla \tilde{h}_q \}$ 
31:     end if
32:      $\Omega_k = \left\langle f^1, \dots, f^{m_u}, \left\{ \begin{matrix} \hat{g}^0, & \dots, & \hat{g}^m \\ g^{m+1}, & \dots, & g^{m_w} \end{matrix} \right\} \middle| \Omega^* \right\rangle$ .
33:     if  $(k-1 \geq \hat{k}^m) \& (\Omega_k == \Omega_{k-1})$  { The highest unknown input degree of reconstructability is  $m < m_w$  (system not canonic), and the observability codistribution is  $\Omega_k$  } then
34:       Set  $\mathcal{O} = \Omega_k$  and set  $\mathcal{E} = \Sigma$ .
35:       Continue=false.
36:       break main and nested loop
37:     end if
38:     if  $\deg_w(\Omega_k) > m$  then
39:       Set  $\Omega = \Omega_k$ .
40:       break nested loop
41:     end if
42:   end while
43: end while
44: if Continue then
45:    $[\tilde{h}_1, \dots, \tilde{h}_{m_w}] = S(\Sigma, \Omega)$ .
46:   Compute  $\mu, \nu, \hat{g}^\alpha$  (Eqs (4)-(5)), and  $\tilde{\mathcal{O}}$  (Eq. (20) in Martinelli (2022)).
47:    $\mathcal{O} = \left\langle f^1, \dots, f^{m_u}, \hat{g}^0, \hat{g}^1, \dots, \hat{g}^{m_w} \middle| \Omega + \tilde{\mathcal{O}} \right\rangle$  and set  $\mathcal{E} = \Sigma$ .
48: end if

```

Algorithm 2: The algorithm that returns the integer s_x^m at Line 19 of Algorithm 1.

```

1: Set  $\Sigma' = \Sigma, m, \tilde{h}_1, \dots, \tilde{h}_m$ , as at Line 11 of Algorithm 1.
2: Set  $k = 0, \Omega_0^m = \text{span} \{ \nabla \tilde{h}_1, \dots, \nabla \tilde{h}_m \}$ , and  ${}^x\Omega_0^m = \Omega_0^m$ .
3: loop
4:   Set  $k = k + 1$ .
5:   Reset  $[\Sigma', \Omega_{k-1}^m] = \mathcal{A}(\Sigma', m, \Omega_{k-1}^m)$ 
6:    $\Omega_k^m = \Omega_{k-1}^m + \mathcal{L}_{g^0}\Omega_{k-1}^m + \sum_{j=1}^m \mathcal{L}_{g^j}\Omega_{k-1}^m$ .
7:    ${}^x\Omega_k^m = D_x(\Omega_k^m)$ .
8:   if  ${}^x\Omega_k^m == {}^x\Omega_{k-1}^m$  then
9:     Set  $s_x^m = k$  then exit.
10:  end if
11: end loop

```

Algorithm 3: The algorithm that returns the integer r^m at Line 19 of Algorithm 1.

```

1: Set  $\Sigma' = \Sigma$  and  $m$  as at Line 11 of Algorithm 1.
2: Set  $k = 0, {}^m\Delta_0 = \text{span} \{ f^1, \dots, f^{m_u} \}$ .
3: loop
4:   Set  $k = k + 1$ .
5:   Reset  $[\Sigma', {}^m\Delta_{k-1}] = \mathcal{A}(\Sigma', m, {}^m\Delta_{k-1})$ 
6:    ${}^m\Delta_k = {}^m\Delta_{k-1} + \sum_{\beta=0}^m {}^m\Delta_{k-1}^{[\beta]}$ 
7:   if  ${}^m\Delta_k == {}^m\Delta_{k-1}$  then
8:     Set  $r^m = k - 1$  then exit.
9:   end if
10: end loop

```

2.2.1. Functioning and outputs of Algorithm 1

Algorithm 1 is recursive and consists of a main loop that may need to run a nested loop. The execution of the nested loop automatically sets up a new system which is a finite unknown input extension of the original system.⁹ In many cases, the nested loop is not executed. When executed, the state augmentation is minimal, in the sense that only the unknown inputs needed to achieve the highest unknown input degree of reconstructability¹⁰ are included. In addition, the highest unknown input degree of reconstructability is always achieved in a finite number of steps.¹¹ Algorithm 1 automatically selects which unknown inputs must be included. In this paper, we denote by $\mathcal{E}$ this special unknown input extension. When the nested loop is not executed, $\mathcal{E}$ could differ from the original system only because its unknown inputs could have been re-ordered and, consequently, the vector fields $g^1, \dots, g^{m_w}$ in (3) exchanged with each other. When the nested loop is executed, $\mathcal{E}$ certainly differs from the original system because some of its unknown inputs have become the time derivatives of a given order of the original unknown inputs.

The execution of Algorithm 1 is based on the following operations: $\deg_w$, S , $\mathcal{R}$, and $\mathcal{A}$. Their complete definition is available in Sections 3.1.1–3.1.4 of Martinelli (2023c), respectively. Here, we summarize these definitions as follows:

⁹ Given the system characterized by (3), a finite unknown input extension of it is obtained by including in the state some of (or all) the unknown inputs, together with their time derivatives up to a given order (the order is not necessarily the same for all the unknown inputs which are included). The resulting extended system is still characterized by (3) and the new functions can be easily computed (see Section 5.1 of Martinelli, 2022 and in particular Equations (22–23) in Martinelli, 2022).

¹⁰ The definitions of the unknown input degree of reconstructability and of the highest unknown input degree of reconstructability are given in Martinelli (2022) (Definitions 2 and 6, respectively).

¹¹ See Proposition 2 and Lemma 1 in Martinelli (2022).

- Given a codistribution, $\deg_w$ computes the degree of reconstructability of the system from the generators of this codistribution.
- Given a codistribution, S selects a set of independent observable functions from the generators of this codistribution that make the degree of reconstructability of the system from them equal to the degree of reconstructability of the system from the generators of that codistribution.
- Given a set of m observable functions such that the unknown input degree of reconstructability from them is m , $\mathcal{R}$ reorders the unknown inputs of the system in such a way that the unknown input reconstructability matrix¹² from the above m observable functions has the first m columns linearly independent.
- Given an integer $m < m_w$, $\mathcal{A}$ provides the unknown input extension that includes the last $m_w - m$ unknown inputs in the state.

In addition, the execution of Algorithm 1 needs to compute minimal distributions (or codistributions) that contains a given distribution (or codistribution) and which are invariant with respect to a set of vector fields. All these minimal distributions (or codistributions) are denoted by angle brackets (e.g., see Line 47 of Algorithm 1). These minimal distributions (or codistributions) can be easily computed by executing simple recursive algorithms, which are available in the literature (e.g., Algorithms (1.39) and (1.45) in Isidori (1995)). The reader can find a quick summary of these recursive algorithms in Section 2.4 of Martinelli (2023c) (where the notation is the same as in this paper). Finally, the symbol $\mathcal{L}$ denotes the Lie derivative operator.

Algorithm 1 provides, in a finite number of steps, the aforementioned unknown input extension $\mathcal{E}$. In addition, it returns the unknown input degree of reconstructability of $\mathcal{E}$, denoted by m , together with a set of observable functions, $\tilde{h}_1, \dots, \tilde{h}_m$, such that the unknown input reconstructability matrix of $\mathcal{E}$ from $\tilde{h}_1, \dots, \tilde{h}_m$ is full rank. Note that if $\mathcal{E}$ is canonic with respect to its unknown inputs then $m = m_w$.¹³ Finally, Algorithm 1 builds the observability codistribution of $\mathcal{E}$, which is denoted by $\mathcal{O}$.

2.2.2. The tensor fields μ_β^α , ν_β^α , and $\hat{g}^\alpha$

The execution of Algorithm 1 needs to compute the following tensor quantities¹⁴: μ_β^α , ν_β^α , and $\hat{g}^\alpha$. The two-index tensor μ is defined as follows:

$$\mu_j^i = \mathcal{L}_{g^i} \tilde{h}_j, \quad i, j = 1, \dots, m \quad (4)$$

$$\mu_0^0 = 1, \quad \mu_i^0 = 0, \quad \mu_i^0 = \frac{\partial \tilde{h}_i}{\partial t} + \mathcal{L}_{g^0} \tilde{h}_i, \quad i = 1, \dots, m.$$

The tensor μ is non singular by construction (see Section 4 in Martinelli (2022), when $m = m_w$, and Sections 5 and 6, when $m < m_w$) and we denote by ν its inverse. In Martinelli (2022), when $m < m_w$, the tensor ν was denoted by ${}^m\nu$. Similarly, the tensor μ was denoted by ${}^m\mu$. Here, we use μ and ν independently of the value of m .

Starting from ν we build the new vector fields $\hat{g}^0, \dots, \hat{g}^m$ defined as follows:

$$\hat{g}^\alpha = \sum_{\beta=0}^m \nu_\beta^\alpha g^\beta, \quad \alpha = 0, 1, \dots, m. \quad (5)$$

In Martinelli (2022), when $m < m_w$, the vector fields $\hat{g}^0, \dots, \hat{g}^m$ were denoted by ${}^m\hat{g}^0, \dots, {}^m\hat{g}^m$.

¹² See Definition 1 in Martinelli (2022) for the definition of the unknown input reconstructability matrix.

¹³ See Definition 3 in Martinelli (2022) for the definition of canonic system with respect to its unknown inputs.

¹⁴ In accordance with the Einstein notation, Latin indices take positive values (1, 2, ...), and Greek indices non negative values (0, 1, 2, ...).

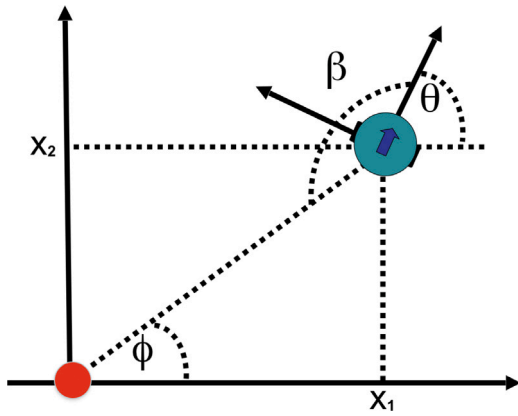


Fig. 1. The body Cartesian coordinates (x_1, x_2) , the orientation (θ) , and the angle measured by the sensor (β) .

2.3. Illustration of Algorithm 1 with an elementary system

We introduce an elementary example of an ODE model that will be used to illustrate the functioning of Algorithm 1 (and later to illustrate further theoretical concepts and the functioning of Algorithm 4). We will refer to this elementary system as the *case study*.

We consider a body that moves in a 2D-environment. The configuration of the body in a global reference frame will be characterized through the vector $[x_1, x_2, \theta]^T$, where x_1 and x_2 are the Cartesian coordinates, and θ is the body orientation (see Fig. 1 for an illustration). We assume that the body is constrained to instantaneously move along the θ direction. On the other hand, the body is also free to rotate. In other words, its dynamics are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ \omega \end{bmatrix} \quad (6)$$

where v and ω are the linear and the rotational speed, respectively, and they are the system inputs. We assume that the body is equipped with a sensor that provides the bearing angle of the origin of the global frame in the body local frame (at the origin there is a landmark perceived by the sensor). In other words, the sensor provides the angle β in Fig. 1. We can express β in terms of the state. We have: $\beta = \pi - (\theta - \phi)$, where $\phi = \arctan\left(\frac{x_2}{x_1}\right)$. We will refer to the two scenarios defined by the following settings:

1. v is known and ω is unknown.
2. Both v and ω are unknown.

We use Algorithm 1 to determine the observability codistribution of these two scenarios.

2.3.1. First scenario

The system is directly a special case of (3). In particular, $x = [x_1, x_2, \theta]^T$, $m_u = 1$, $m_w = 1$, $p = 1$, $h_1(x) = \phi - \theta = \arctan\left(\frac{x_2}{x_1}\right) - \theta$ (we ignore the constant term π in β , which does not affect the observability of the state), and

$$g^0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, g^1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, f^1 = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} \quad (7)$$

Let us run Algorithm 1. Line 2 provides $\Omega = \text{span} \{ \nabla h_1 \} = \text{span} \left\{ \left[-\frac{x_2}{x_1^2 + x_2^2}, \frac{x_1}{x_1^2 + x_2^2}, -1 \right] \right\}$. The execution of Algorithm 1 continues with the execution of the main loop. Line 5 requires to compute $\deg_{\mathcal{M}}(\Omega)$. As we aforementioned, the operation $\deg_{\mathcal{M}}$ is defined in

Section 3.1.1 of [Martinelli \(2023c\)](#). In this case, we need to compute the Lie derivative of h_1 (i.e., the generator of Ω) along g^1 . If the result does not vanish, $\deg_w(\Omega) = 1$. We have: $\mathcal{L}_{g^1} h_1 = \nabla h_1 \cdot g^1 = -1 \neq 0$. As a result, $\deg_w(\Omega) = 1$ ($= m_w$). Hence, the condition at Line 5 is met and the main loop terminates. The system is in canonical form with respect to its unknown input. The execution of Algorithm 1 continues with Line 45. The operation S is defined in Section 3.1.1 of [Martinelli \(2023c\)](#). In this case, we have: $\tilde{h}_1 = S(\Sigma, \Omega) = h_1$. Line 46 provides $\mu, \nu, \hat{g}^0, \hat{g}^1$, and $\tilde{\Omega}$. The tensor μ is obtained from (4). We have:

$$\mu_0^0 = 1, \mu_0^1 = \mu_1^0 = 0, \mu_1^1 = -1, \text{ and } \nu = \mu, \quad (8)$$

(the tensor ν is the inverse of μ and, in this special case, the two tensors coincide). From (5), we obtain:

$$\hat{g}^0 = v_0^0 g^0 + v_1^0 g^1 = g^0,$$

$$\hat{g}^1 = v_0^1 g^0 + v_1^1 g^1 = -g^1$$

Finally, by using Equation (20) in [Martinelli \(2022\)](#), we have

$$\tilde{\mathcal{O}} = \text{span} \{ \nabla h_2, \nabla h_3 \}$$

with $h_2 = \mathcal{L}_{f^1} \tilde{h}_1 = \frac{\sin(\theta-\phi)}{\rho}$, $h_3 = \mathcal{L}_{[f^1](1)} \tilde{h}_1 = -\frac{\cos(\theta-\phi)}{\rho}$, and $\rho = \sqrt{x^2 + y^2}$.

Note that the covectors ∇h_1 , ∇h_2 , and ∇h_3 are not independent, and, consequently, $\Omega + \tilde{\mathcal{O}} = \text{span} \{ \nabla h_1, \nabla h_2 \}$.

The last step of Algorithm 1 is the execution of Line 47. This computes the minimal codistribution that contains $\Omega + \tilde{\mathcal{O}}$ and that is invariant under the Lie derivatives along $\hat{g}^0 = g^0$, $\hat{g}^1 = -g^1$, and f^1 , simultaneously. This is immediately obtained by running standard recursive algorithms (e.g., the algorithm in Eq. (1.45) of [Isidori \(1995\)](#), or the algorithm in Eq. (2.8) of [Martinelli \(2023c\)](#)). In this specific case we obtain that the codistribution $\Omega + \tilde{\mathcal{O}}$ is invariant under the Lie derivatives along $\hat{g}^0 = g^0$, $\hat{g}^1 = -g^1$, and f^1 . We finally obtain:

$$\mathcal{O} = \text{span} \{ \nabla h_1, \nabla h_2 \}. \quad (9)$$

Note that, for this system, Algorithm 1 did not execute the nested loop. As a result, the system $\mathcal{E}$ returned by Algorithm 1 coincides with the original system. We conclude by listing the outputs computed by Algorithm 1 for this system: (i) the system $\mathcal{E}$ is the original system, (ii) the observability codistribution is given in (9), (iii) the unknown input degree of reconstructability is $m = 1 = m_w$ (system canonic with respect to its UI), and (iv) $\tilde{h}_1 = h_1 = \arctan\left(\frac{x_2}{x_1}\right) - \theta$.

In Section 2.4.1, we discuss the observability properties that can be obtained from the structure of $\mathcal{O}$ in (9). In particular, we show that these properties agree with our intuition.

2.3.2. Second scenario

The system is directly a special case of (3). In particular, $x = [x_1, x_2, \theta]^T$, $m_u = 0$, $m_w = 2$, $p = 1$, $h_1(x) = \phi - \theta$, and

$$g^0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, g^1 = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix}, g^2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Let us run Algorithm 1. Line 2 provides $\Omega = \text{span}\{\nabla h_1\}$, as in the previous case. The execution of Algorithm 1 continues with the execution of the main loop. Line 5 requires to compute $\deg_w(\Omega)$. We have: $\mathcal{L}_{g^*} h_1 = \nabla h_1 \cdot g^* = \frac{\sin(\theta-\phi)}{\rho} \neq 0$. As a result, $\deg_w(\Omega) = 1 (< m_w)$. Hence, the condition at Line 5 is not met and the loop at Lines 9–14 is executed. Lines 10–12 provide $m = 1$, $\tilde{h}_1 = h_1 = \phi - \theta$, $\mu_0^0 = 1$, $\mu_1^0 = \mu_1^0 = 0$, $\mu_1^1 = \frac{\sin(\theta-\phi)}{\rho}$, $v_0^0 = 1$, $v_0^1 = v_0^1 = 0$, $v_1^1 = \frac{\rho}{\sin(\theta-\phi)}$.

$$g^0 = v_0^0 g^0 + v_1^0 g^1 = g^0,$$

$$\hat{g}^1 = v_0^1 g^0 + v_1^1 g^1 = \frac{\rho}{\sin(\theta - \phi)} g^1.$$

On the other hand, Line 13 does not determine new observable functions, as $\mathcal{L}_{g_2} h_1 = -1 \neq 0$. Hence, $\deg_w(\Omega) = 1 = m$ and the loop at Lines 9–14 is not repeated. In addition, as $m_u = 0$, Line 21 sets $\hat{k}^m = 0$ and Line 23 set $k = 0$ and $\Omega_0 = \Omega (= \text{span}\{\nabla h_1\})$. Now, the nested loop is executed and the state is augmented by including the second UI. The new state is:

$$x = [x_1, x_2, \theta, \omega]^T \quad (10)$$

The augmented system is characterized by the following vector fields:

$$g^0 = \begin{bmatrix} 0 \\ 0 \\ \omega \\ 0 \end{bmatrix}, g^1 = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \\ 0 \end{bmatrix}, g^2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (11)$$

Line 27 provides

$$\mu_0^0 = 1, \mu_1^0 = -\omega, \mu_0^1 = 0, \mu_1^1 = \frac{\sin(\theta - \phi)}{\rho}, \quad (12)$$

$$v_0^0 = 1, v_1^0 = \frac{\rho\omega}{\sin(\theta - \phi)}, v_0^1 = 0, v_1^1 = \frac{\rho}{\sin(\theta - \phi)},$$

$$\hat{g}^0 = v_0^0 g^0 + v_1^0 g^1 = g^0 + \frac{\rho\omega}{\sin(\theta - \phi)} g^1,$$

$$\hat{g}^1 = v_0^1 g^0 + v_1^1 g^1 = \frac{\rho}{\sin(\theta - \phi)} g^1.$$

Line 28 sets $\Omega^* = \Omega_0$ and Line 32 returns $\Omega_1 = \Omega^* = \Omega_0$. As a result, the condition at Line 33 is met and both the main and the nested loop are interrupted. The highest unknown input degree of reconstructability of the system is $m = 1 < m_w = 2$. The system is not canonic with respect to its UIs. The observability codistribution is:

$$\mathcal{O} = \text{span}\{\nabla h_1\}. \quad (13)$$

As for the first scenario, in Section 2.4.2 we discuss the observability properties that can be obtained from the structure of $\mathcal{O}$ in (13) and we show that also in this case these properties agree with our intuition.

2.4. Unobservability of the state, state symmetries and indistinguishable states

In this section we use the case study to illustrate the following fundamental concept. State unobservability is always the consequence of the presence of a system invariance under one or more *continuous transformations*. The interested reader may find in Section 4.6 of Martinelli (2020) more details, which include a quantitative characterization of the aforementioned *continuous transformations*.¹⁵ We will first illustrate this concept by qualitatively discussing a trivial example (that is even simpler than our case study).

Let us suppose that we are moving along a hallway, which is perfectly uniform (i.e., it is invariant under a shift along the axis of the hallway). Intuitively, we know that, after a while, we will definitely get lost along the axis of the hallway. In other words, the state that characterizes our position and orientation is certainly unobservable. In particular, the component of this state that characterizes the coordinate of the position along the axis of the hallway cannot be reconstructed from the sensor measurements. The reason of this feature is that the hallway appears uniform to our sensors. We want to show that we can analytically (and automatically) build the above system invariance directly from the observability codistribution $\mathcal{O}$ returned by Algorithm 1. In particular, this invariance, that we call system symmetry, is characterized by the killing vectors of $\mathcal{O}$. More in general, given any system, the set of all the symmetries is the set of all the independent vectors that belong to the orthogonal distribution $\mathcal{O}^\perp$. A proof of this

fundamental result is available in Section 4.6 of Martinelli (2020). Here, we provide a quick sketch of the proof and then we illustrate this result through our case study.

Let us suppose that the observability codistribution of a given system is $\mathcal{O}$. First of all, if the state is observable the dimension of $\mathcal{O}$ is equal to the dimension of the state. As a result, the orthogonal distribution $\mathcal{O}^\perp$ is empty. Hence, the observability of the state is equivalent to the absence of non trivial killing vectors of $\mathcal{O}$. Now, let us suppose that the state is unobservable and let us denote by ξ a vector that belongs to the orthogonal distribution (i.e., $\xi \in \mathcal{O}^\perp$). We claim that the state x is indistinguishable from $x + \epsilon\xi$, where ϵ is an infinitesimal. In other words, we consider the infinitesimal transformation:

$$x \rightarrow x' = x + \epsilon\xi \quad (14)$$

and we claim that x is indistinguishable from x' .

All the knowledge that we have on the state is equivalent to the knowledge of the values taken by all the observable functions. Let us consider a given observable function and let us denote it by $h(x)$. By definition, we know that $\nabla h \in \mathcal{O}$ (we remind the reader that this is precisely the definition of the observability codistribution). As a result, $\nabla h \cdot \xi = 0$. Let us compute the value of h on the transformed state, i.e., let us compute $h(x')$. We have:

$$h(x') = h(x + \epsilon\xi) \simeq h(x) + \epsilon\nabla h \cdot \xi = h(x)$$

Hence, any observable function takes the same value on the transformed state, meaning that x' and x are indistinguishable.

Clearly, we can apply the infinitesimal transformation in (14) many consecutive times and we obtain a state which is always indistinguishable from the original state. Each application of the infinitesimal transformation in (14) provides a new state that differs from the previous state by $\Delta x' = \epsilon\xi$. Note that, in general, ξ depends on the state (e.g., see later when we compute the symmetries of our case study). By summing up all the changes $\Delta x'$ we obtain a new state which is the solution of the following first order differential equation:

$$\begin{cases} \frac{dx'}{d\tau} = \xi(x'(\tau)) \\ x'(0) = x \end{cases} \quad (15)$$

We illustrate this result by using our case study. We analyse the two scenarios separately.

2.4.1. First scenario

The distribution orthogonal to the codistribution in (9) is:

$$\mathcal{O}^\perp = \text{span}\left\{\begin{bmatrix} -x_2 \\ x_1 \\ 1 \end{bmatrix}\right\} \quad (16)$$

The infinitesimal transformation in (14) becomes:

$$\begin{bmatrix} x_1 \\ x_2 \\ \theta \end{bmatrix} \rightarrow \begin{bmatrix} x'_1 \\ x'_2 \\ \theta' \end{bmatrix} = \begin{bmatrix} x_1 - \epsilon x_2 \\ x_2 + \epsilon x_1 \\ \theta + \epsilon \end{bmatrix},$$

which is an infinitesimal rotation around the vertical axis. The differential equations in (15) become:

$$\begin{cases} \frac{dx'_1}{d\tau} = -x'_2 \\ \frac{dx'_2}{d\tau} = x'_1 \\ \frac{d\theta'}{d\tau} = 1 \\ x'_1(0) = x_1, x'_2(0) = x_2, \theta'(0) = \theta \end{cases}$$

The solution is:

$$\begin{cases} x'_1(\tau) = \cos \tau x_1 - \sin \tau x_2 \\ x'_2(\tau) = \sin \tau x_1 + \cos \tau x_2 \\ \theta'(\tau) = \theta + \tau \end{cases} \quad (17)$$

¹⁵ In particular, in Section 4.6 of Martinelli (2020) we defined these continuous transformations as the result of the action of a Lie group.

which is a rotation of magnitude τ around the vertical axis.

We conclude that the generator of the above $\mathcal{O}^\perp$ expresses the invariance of the system under any rotation around the vertical axis. This result agrees with our intuition. All the available measurements, which provide the linear speed v and the bearing angle β , are independent of the orientation of the global frame in Fig. 1.

2.4.2. Second scenario

The distribution orthogonal to the codistribution in (13) is:

$$\mathcal{O}^\perp = \text{span} \left\{ \begin{bmatrix} -x_2 \\ x_1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad (18)$$

The presence of the first generator is not surprising. By removing the knowledge of the linear speed v , we loose information and the invariance under rotations around the vertical axis certainly remains. The third generator expresses the unobservability of the new component of the state, i.e., the angular speed ω . Let us study the second generator. For the brevity sake, we refer to the first three components of the state (i.e., we refer to the original state). The infinitesimal transformation in (14) becomes, in this specific case:

$$\begin{bmatrix} x_1 \\ x_2 \\ \theta \end{bmatrix} \rightarrow \begin{bmatrix} x'_1 \\ x'_2 \\ \theta' \end{bmatrix} = \begin{bmatrix} x_1 + \epsilon x_1 \\ x_2 + \epsilon x_2 \\ \theta \end{bmatrix},$$

which is an infinitesimal change of the scale. The system in (15) becomes:

$$\begin{cases} \frac{dx'_1}{d\tau} = x'_1 \\ \frac{dx'_2}{d\tau} = x'_2 \\ \frac{d\theta'}{d\tau} = 0 \\ x'_1(0) = x_1, x'_2(0) = x_2, \theta'(0) = \theta \end{cases}$$

Its solution is:

$$\begin{cases} x'_1(\tau) = e^\tau x_1 \\ x'_2(\tau) = e^\tau x_2 \\ \theta'(\tau) = \theta \end{cases}$$

which is a scale transformation. In particular, the scale factor is e^τ . Therefore, the second generator of the above $\mathcal{O}^\perp$ expresses the invariance of the system under any scale transformation. This result agrees with our intuition. By losing the knowledge of v , the system loses the only source of metric information (the available sensor only provides angular measurements).

We conclude by emphasizing that, for both the scenarios, the physical meaning of the symmetries is evident (rotations and scale transformations). This is a lucky case (and this is one of the reasons why we selected this elementary example as a case study). The physical interpretation of the symmetries, which are always algebraically determined by computing the orthogonal distribution, is not always possible (e.g., see later when we compute them in our HIV models). On the other hand, they always allow us to build indistinguishable states (and also indistinguishable unknown inputs, as we will see in Sections 2.6, and 2.7). In addition, they also allow us to characterize the missing information to make observable the state and identifiable all the parameters, as we will see in Section 2.9.

2.5. Procedure for parameter identifiability: Algorithm 4

Algorithm 4 is a recursive algorithm that provides the local identifiability of all the parameters, constant and time varying, that characterize the system $\mathcal{E}$. Note that the system $\mathcal{E}$ is built by Algorithm

1 (the first instruction of Algorithm 4, i.e., Line 1, executes Algorithm 1) and may differ from the original system because some of the time varying parameters, together with their time derivatives up to a given order, were included in the state ($\mathcal{E}$ may also differ from the original system because its unknown inputs could have been re-ordered). On the other hand, even if this is the case, it is easy to obtain the local identifiability of the original parameters as soon as we have obtained the identifiability of the parameters that characterize $\mathcal{E}$. First, the constant parameters of $\mathcal{E}$ always coincide with the constant parameters of the original system and they are some of the components of the state for both the systems. Let us consider the time varying parameters and let us suppose that the k th time varying parameter, together with its first $\mathcal{K} - 1$ time derivatives, were included in the state of $\mathcal{E}$. Algorithm 4 actually provides the local identifiability of the $\mathcal{K}^{th}$ order time derivative of w_k (i.e., $w_k^{(\mathcal{K})} := \frac{d^{\mathcal{K}} w_k}{dt^{\mathcal{K}}}$). On the other hand, the identifiability of the original time varying parameter and its first $\mathcal{K} - 1$ time derivatives is trivially checked by verifying if their gradients belong to the observability codistribution $\mathcal{O}$, which is provided by Algorithm 1. In other words, it suffices to verify the condition $\nabla w_k^{(i)} \in \mathcal{O}$, for $i = 0, 1, \dots, \mathcal{K} - 1$.

Algorithm 4: The algorithm that provides the local identifiability of all the system parameters (constant and time varying). Comments are in blue.

```

1: Compute  $\mathcal{E}, \mathcal{O}, m, \tilde{h}_1, \dots, \tilde{h}_m$ , by running Algorithm 1.
2: for  $i = 1 : q$  { This loop checks the identifiability of the constant parameters } do
3:   if  $\nabla Q_i \in \mathcal{O}$  then
4:      $Q_i$  is locally identifiable
5:   else
6:      $Q_i$  is not locally identifiable
7:   end if
8: end for
9: if  $m < m_w$  { Identifiability of the last  $m_w - m$  time varying parameters } then
10:   $w_{m+1}, \dots, w_{m_w}$ , are not locally identifiable.
11: end if
12: Compute  $\mu$  from (4).
13: Compute  $\nu$  (the inverse of  $\mu$ ).
14: Compute  $\mathcal{O}^\perp$  and set  $\mathcal{O}^\perp = \text{span}\{\xi^1, \dots, \xi^S\}$ .
15: for  $i_s = 1 : S$  do
16:   Set  $\xi = \xi^{i_s}$ .
17:   Compute  $\xi_j^\alpha$  ( $\alpha = 0, 1, \dots, m$  and  $i = 1, \dots, m$ ) from (19), and set  $\Xi_i^{\alpha, i_s} = \xi_j^\alpha$ .
18: end for
19: for  $j = 1 : m$  { This loop checks the identifiability of the first  $m$  time varying parameters } do
20:   if  $C_j^{\alpha, i_s} == 0, \forall \alpha = 0, 1, \dots, m, \forall i_s = 1, \dots, S$  {  $C_j^{\alpha, i_s}$  is defined by Eq. (20) } then
21:      $w_j$  is locally identifiable
22:   else
23:      $w_j$  is not locally identifiable
24:   end if
25: end for
```

2.5.1. Description of Algorithm 4

The first instruction (Line 1) executes Algorithm 1. This returns the outputs described at the end of Section 2.2.1, i.e., $\mathcal{E}, m, \tilde{h}_1, \dots, \tilde{h}_m$, and $\mathcal{O}$.

The first loop of Algorithm 4 (Lines 2–8) checks the local identifiability of all the constant parameters.

The rest of the Algorithm checks the local identifiability of the time varying parameters that characterize $\mathcal{E}$.

If the system $\mathcal{E}$ is not canonic with respect to its unknown inputs (i.e., $m < m_w$), the last $m_w - m$ time varying parameters of $\mathcal{E}$ are not identifiable, even locally (Lines 9–11). This result can be obtained starting from the results given in Sections 7 and 8 of Martinelli (2022). The reader can find a direct proof of this result in Martinelli (2023b) (first part of Theorem 3).

Then, Algorithm 4 checks the local identifiability of the first m time varying parameters of $\mathcal{E}$. Lines 12 and 13 compute the $(1, 1)$ -tensor fields μ and ν , respectively. Line 14 computes the distribution orthogonal to $\mathcal{O}$, denoted by $\mathcal{O}^\perp$, and a set of its generators. They are denoted by $\xi^1, \dots, \xi^S$ (S is the dimension of this orthogonal distribution).

Then, for each generator of $\mathcal{O}^\perp$, the loop at Lines 15–18 computes the following quantities, for any $\alpha = 0, 1, \dots, m$ and any $i = 1, \dots, m$:

$$\xi_i^\alpha = \begin{cases} \left(\frac{\partial}{\partial x} \tilde{\mathcal{L}}_{g^0} \tilde{h}_i \right) \cdot \xi, & \alpha = 0 \\ \left(\frac{\partial}{\partial x} \mathcal{L}_{g^\alpha} \tilde{h}_i \right) \cdot \xi, & \alpha = 1, \dots, m \end{cases} \quad (19)$$

with $\tilde{\mathcal{L}}_{g^0} = \mathcal{L}_{g^0} + \frac{\partial}{\partial t}$.

Note that the above quantities (ξ_i^α) depend on the selected generator (ξ in (19)) of $\mathcal{O}^\perp$. All these quantities are stocked in Ξ_i^{α, i_s} , where the index i_s specifies the selected generator of $\mathcal{O}^\perp$.

The loop at Lines 19–25 checks the identifiability of the first m time varying parameters of $\mathcal{E}$. Specifically, the j th time varying parameter (w_j) is locally identifiable if and only if all the quantities

$$C_j^{\alpha, i_s} = \sum_{i=1}^m v_i^j \Xi_i^{\alpha, i_s} \quad (20)$$

$\forall \alpha = 0, 1, \dots, m, \forall i_s = 1, \dots, S$ vanish, simultaneously. It suffices that one of the above quantities does not vanish (i.e., there exists a value of α and a value of i_s such that $C_j^{\alpha, i_s} \neq 0$) to conclude that w_j is not locally identifiable. This result can be obtained starting from the results given in Sections 7 and 8 of Martinelli (2022). A direct proof is available in Martinelli (2023b) (first part of Theorem 2).

2.6. Symmetries of the unknown input vector

In this section we extend the analysis carried out in Section 2.4 to the unknown inputs and we provide the expression of all the symmetries that characterize the unknown input vector. In other words, we provide the expression of all the vectors $\chi = [\chi_1, \dots, \chi_{m_w}]^T$ such that:

$$w' = w + \epsilon \chi \quad (21)$$

is indistinguishable from w .

For the clarity sake, we provide an explanation about the meaning of indistinguishability of the unknown input. As we will see, it is intimately related to the indistinguishability of the state or to the system non canonicity.

Let us assume that the initial time is $t = 0$ and the initial state is $x(0) = x_0$. The state at time t is obtained by solving the following differential equation:

$$\begin{cases} \dot{x}(t) = g^0(x, t) + \sum_{k=1}^{m_u} f^k(x, t) u_k(t) + \sum_{j=1}^{m_w} g^j(x, t) w_j(t) \\ x(0) = x_0 \end{cases} \quad (22)$$

We introduce the state $x'(t)$ obtained by solving the following differential equation:

$$\begin{cases} \dot{x}'(t) = g^0(x', t) + \sum_{k=1}^{m_u} f^k(x', t) u_k(t) + \sum_{j=1}^{m_w} g^j(x', t) w'_j(t) \\ x'(0) = x'_0 \end{cases} \quad (23)$$

We emphasize that the state $x'(t)$ is obtained by solving the same differential equation that provides $x(t)$ but with the following two differences:

- The initial state is x'_0 instead of x_0 .
- The unknown input is $w'(t) = [w'_1(t), \dots, w'_{m_w}(t)]^T$ instead of $w(t) = [w_1(t), \dots, w_{m_w}(t)]^T$ (and this second difference involves the unknown input at any time in the considered time interval and not simply the initial time).

We are interested in determining all the initial states x'_0 and all the unknown input functions $w'(t)$ such that, independently of the choice of the known input functions (i.e., $u_1(t), \dots, u_{m_u}(t)$), the states $x'(t)$ produce exactly the same outputs of the states $x(t)$, i.e.:

$$h_i(x(t), t) = h_i(x'(t), t), \quad i = 1, \dots, p,$$

for any $t \in [0, T]$.

In this section we restrict the analysis to infinitesimal transformations and we determine the expression of the vector χ in (21). In Section 2.7 we extend the results to the case of finite transformations.

We have two distinct sets of symmetries. The first set is not empty if and only if the system is not canonic with respect to its unknown inputs (i.e., $m < m_w$). They will be denoted by ${}^c\chi$, where c stands for *system non-Canonicity*. The second set may be not empty when the state is unobservable. They will be denoted by ${}^u\chi$, where u stands for *state Unobservability*. The expression of the symmetries of the first set is:

$${}^c\chi^i = \begin{bmatrix} 0_m \\ e_i \end{bmatrix}, \quad i = 1, \dots, m_w - m, \quad (24)$$

with 0_m the zero m -column vector and e_i the unit vector of dimension $m_w - m$ with zero everywhere except the i th component which is 1. The above vectors are $m_w - m$ independent symmetries of the unknown input vector. This result can be obtained starting from the results given in Sections 7 and 8 of Martinelli (2022). A direct proof is available in Appendix A of Martinelli (2023c). Note that these symmetries only regard the unknown inputs and, as we aforementioned, they exist if and only if the system is not canonic with respect to its unknown inputs ($m < m_w$). From them we obtain the following first set of infinitesimal transformations:

$$x'_0 = x_0, \quad w'(t) = w(t) + \epsilon {}^c\chi^i, \quad i = 1, \dots, m_w - m \quad (25)$$

In other words, by using the above x'_0 and $w'(t)$, the solution $x'(t)$ of (23) provides the same outputs of the solution $x(t)$ of (22). This means that the above x'_0 and $w'(t)$ cannot be distinguished from x_0 and $w(t)$.

The expression of the symmetries of the second set is:

$$\begin{cases} {}^u\chi_k = -\sum_{i=1}^m v_k^i \left(\xi_i^0 + \sum_{j=1}^m \xi_i^j w_j \right) & k = 1, \dots, m \\ {}^u\chi_k = 0 & k = m + 1, \dots, m_w \end{cases} \quad (26)$$

with ξ_i^α given in (19), where we use any $\xi \in \mathcal{O}^\perp$ (symmetry of the state). This result can be obtained starting from the results given in Sections 7 and 8 of Martinelli (2022). A direct proof is available in Appendix B of Martinelli (2023c). Note that these symmetries regard the state and the unknown inputs, simultaneously. Additionally, given a symmetry ξ of the state, the corresponding symmetry ${}^u\chi$ in (26) can be trivial, i.e., can be a null vector. This is certainly the case when all the quantities ξ_i^α in (19) vanish. This means that there are cases where the first m unknown inputs can be reconstructed even in the presence of state unobservability. Hence, it is better to call the vector ${}^u\chi$ in (26) a potential symmetry of the unknown input vector. We have a potential symmetry of the unknown input vector for any symmetry of the state.

From the symmetries in (26) we obtain the following second set of infinitesimal transformations:

$$x'_0 = x_0 + \epsilon \xi^i, \quad w'(t) = w(t) + \epsilon {}^u\chi^i, \quad i = 1, \dots, S \quad (27)$$

where $\xi^i \in \mathcal{O}^\perp$ is the symmetry of the state that provides ${}^u\chi^i$, in accordance with (26) (i.e., ${}^u\chi^i$ is obtained by setting $\xi = \xi^i$ in (19)). Again, by using the above x'_0 and $w'(t)$, the solution $x'(t)$ of (23) provides the same outputs of the solution $x(t)$ of (22). In other words, the above x'_0 and $w'(t)$ cannot be distinguished from x_0 and $w(t)$.

2.7. Indistinguishable states and indistinguishable unknown inputs

In this section we want to extend the results obtained in Section 2.6 to the case of finite transformations. This is precisely what we carried out in Section 2.4 to determine the differential equation in (15) starting from the infinitesimal transformation given in (14).

Let us consider the infinitesimal transformation given in (25). By repeating it many times we obtain the following result. First of all, the transformation does not affect x_0 and we trivially obtain $x'_0 = x_0$. Regarding the unknown inputs, we have $m_w - m$ independent $w' = w'(t, \tau)$, which are the solutions of the following differential equations:

$$\begin{cases} \frac{dw'}{d\tau} = {}^c\chi^i \\ w'(t, 0) = w(t), \end{cases}$$

$i = 1, \dots, m_w - m$. In the above equation, the time t is a fixed parameter. In other words, to obtain $w'(t, \tau)$, we first set the time to the desired value t . Due to the trivial expression of ${}^c\chi$ in (24) (and in particular its independence of the specific case), this differential equation can be solved once forever and provides:

$$\begin{cases} w'_j(t, \tau) = w_j(t), & j \neq m + i \\ w'_j(t, \tau) = w_j(t) + \tau, & j = m + i \end{cases} \quad (28)$$

For the S infinitesimal transformations given in (27), the corresponding finite transformations now involve simultaneously the initial state and the unknown inputs. To analytically compute the corresponding finite transformations, we need the following condition to be satisfied:

$$\left[\xi, g^0 + \sum_{i=1}^{m_u} f^i u_i + \sum_{j=1}^{m_w} g^j w_j \right] + \sum_{j=1}^m g^j {}^u\chi_j = 0, \quad (29)$$

where the square brackets are the Lie brackets. This condition expresses the commutativity between the temporal shift and the shift generated by the considered symmetry.

If the condition in (29) is met, the result of many consecutive applications of the S infinitesimal transformations in (27) can be analytically computed. Specifically, for the i th transformation in (27) we obtain the new $x'_0 = x'_0(\tau)$, and $w' = w'(t, \tau)$ by solving the following differential equations:

$$\begin{cases} \frac{dx'}{d\tau} = \xi^i(x'(t, \tau)) \\ \frac{dw'}{d\tau} = {}^u\chi^i(x'(t, \tau), w'(t, \tau)) \\ x'(t, 0) = x(t), \quad w'(t, 0) = w(t), \end{cases} \quad (30)$$

$i = 1, \dots, S$. Again, in the above equation, the time t is a fixed parameter. In other words, to obtain $x'(t, \tau)$, and $w'(t, \tau)$, we first set the time to the desired value t . If, for any $t \in [0, T]$, the above equation admits a solution in the interval $\mathcal{U}$, with $\mathcal{U}$ an open interval of $\mathbb{R}$ that includes $\tau = 0$, then, for any $\tau \in \mathcal{U}$, the unknown input vector $w'(t, \tau)$ is indistinguishable from the unknown input vector $w(t)$. Note that, in the above equation, the value of $x(t)$, which appears at the initial condition $x'(t, 0) = x(t)$, is the solution of (22) at the time t . From $x'(t, \tau)$ we obtain the finite transformation of the initial state, which is the value of this function at the initial time, i.e., $x'_0 = x'_0(\tau) = x'(0, \tau)$.

The solution of (30) is obtained by first determining $x'(t, \tau)$ (i.e., by first solving $\frac{dx'}{d\tau} = \xi^i(x'(t, \tau))$, with initial condition $x'(t, 0) = x(t)$), and then by using it in $\frac{dw'}{d\tau} = {}^u\chi^i(x'(t, \tau), w'(t, \tau))$, which becomes a differential equation of only $w'(t, \tau)$. Note that the solution of $\frac{dw'}{d\tau} = \xi^i(x'(t, \tau))$, with initial condition $x'(t, 0) = x(t)$, is precisely the solution of (15) with $x = x(t)$ and for the same state symmetry (i.e., $\xi = \xi^i$).

2.8. Illustration by the case study

We illustrate the functioning of Algorithm 4 and the theoretical concepts introduced in Sections 2.6 and 2.7 by applying them to our case study. We study separately the two scenarios of our case study.

2.8.1. Identifiability analysis — first scenario

We use Algorithm 4 to obtain the identifiability of the time varying parameter $\omega(t)$.

As the system does not contain constant parameters, the first loop (Lines 2–8) is skipped.

The condition at Line 9 is not satisfied (the system is canonic and $m = m_w = 1$). The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (8). Line 14 provides the distribution given in (16).

The loop at Lines 15–18 consists of the single value $i_s = 1$. For this value, i.e., for the generator of $(\mathcal{O})^\perp$ in (16), we compute the coefficients ξ_i^α , for $\alpha = 0, 1$, and $i = 1$, by using (19), (7), (16), and $\tilde{h}_1 = \phi - \theta$. We have: $\mathcal{L}_{g^0}\tilde{h}_1 = 0$, $\mathcal{L}_{g^1}\tilde{h}_1 = -1$, and $\Xi_1^{0,1} = \xi_1^0 = 0$, $\Xi_1^{1,1} = \xi_1^1 = 0$. As a result, the quantities in (20) vanish for any $\alpha = 0, 1$. Hence, the condition at Line 20 is met, meaning that the time varying parameter $\omega(t)$ is locally identifiable.

2.8.2. Indistinguishable states and parameters — first scenario

As there are no unidentifiable parameters, we only have indistinguishable states that are the ones already determined in Section 2.4.1, given in (17).

2.8.3. Identifiability analysis — second scenario

We use Algorithm 4 to obtain the identifiability of the system returned by Algorithm 1, which is characterized by the augmented state in (10). Note that the time varying parameters are $v(t)$ and $\dot{\omega}(t)$. Regarding the original time varying parameter $\omega(t)$, it belongs to the state of $\mathcal{E}$ and we already obtained its unidentifiability in Section 2.4.2 (its gradient is not orthogonal to the third generator of the distribution in (18) and, consequently, does not belong to the observability codistribution).

As the system does not contain constant parameters, the first loop (Lines 2–8) is skipped.

The condition at Line 9 is satisfied (the system is not canonic and $m = 1 < m_w = 2$). We conclude that $\dot{\omega}(t)$ is unidentifiable, even locally.

The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (12). Line 14 provides the distribution given in (18).

The loop at Lines 15–18 consists of the three values $i_s = 1, 2, 3$ (the distribution $(\mathcal{O})^\perp$ in (18) has three generators). From (11) and by knowing that $\tilde{h}_1 = \phi - \theta$ we have: $\mathcal{L}_{g^0}\tilde{h}_1 = -\omega$. Hence, for $i_s = 3$, i.e., by using for ξ the third generator of the distribution in (18), we obtain: $\xi_1^0 = \frac{\partial \mathcal{L}_{g^0}\tilde{h}_1}{\partial x} \xi = -1$. Hence, $\Xi_1^{0,3} = -1$, and, from (20), $C_1^{0,3} = v_1^1 \Xi_1^{0,3} = -\frac{\rho}{\sin(\theta - \phi)} \neq 0$, where we used the expression in (12) for v_1^1 . Therefore, the condition at Line 20 is not met for $\alpha = 0$ and $i_s = 3$, meaning that the time varying parameter $v(t)$ is unidentifiable, even locally.

2.8.4. Indistinguishable states and parameters — second scenario

We know that the unknown input degree of reconstructability is $m = 1$. As $m_w = 2$, in accordance with (24), we have a single symmetry of the unknown input vector due to the system non canonicity. In particular, we have ${}^c\chi^1 = [0, 1]^T$. From (28), we obtain that all the time varying parameters $\dot{\omega}'(t, \tau) = \dot{\omega}(t) + \tau$ are indistinguishable from $\dot{\omega}(t)$, for any $\tau \in \mathbb{R}$.

Let us determine indistinguishable values of the first time varying parameter $v(t)$. They are related to the unobservability of the state. We need to solve the differential equations in (30). We first need to compute the symmetries of the unknown input vector that correspond to the three generators of $\mathcal{O}^\perp$ in (18). In other words, we compute ${}^u\chi$ in (26). We start by computing the coefficients ξ_1^0 and ξ_1^1 for the three generators.

We have:

$$\mathcal{L}_{g^0}\tilde{h}_1 = -\omega, \quad \mathcal{L}_{g^1}\tilde{h}_1 = \frac{\sin(\theta - \phi)}{\rho}.$$

For the first generator of $\mathcal{O}^\perp$ in (18) we obtain

$$\xi_1^0 = \frac{\partial \mathcal{L}_{g^0} \tilde{h}_1}{\partial x} \xi = 0, \quad \xi_1^1 = \frac{\partial \mathcal{L}_{g^1} \tilde{h}_1}{\partial x} \xi = 0.$$

From (26) we obtain that the corresponding ${}^u\chi$ vanishes.

Let us consider the second generator of $\mathcal{O}^\perp$ in (18). We obtain

$$\xi_1^0 = \frac{\partial \mathcal{L}_{g^0} \tilde{h}_1}{\partial x} \xi = 0, \quad \xi_1^1 = \frac{\partial \mathcal{L}_{g^1} \tilde{h}_1}{\partial x} \xi = -\frac{\sin(\theta - \phi)}{\rho}.$$

From (26) we obtain the corresponding ${}^u\chi$:

$${}^u\chi_1 = -v_1^1 \xi_1^1 w_1 = \frac{\rho}{\sin(\theta - \phi)} \frac{\sin(\theta - \phi)}{\rho} v = v, \quad {}^u\chi_2 = 0,$$

where we used the expression of v_1^1 in (12).

By an explicit computation, it is possible to check that the condition in (29) is met. The differential equations system in (30) becomes:

$$\begin{cases} \frac{dx_1'}{d\tau} = x_1'(t, \tau) \\ \frac{dx_2'}{d\tau} = x_2'(t, \tau) \\ \frac{d\theta'}{d\tau} = 0 \\ \frac{d\omega'}{d\tau} = 0 \\ \frac{dv'}{d\tau} = v'(t, \tau) \\ x_1'(t, 0) = x_1(t), \quad x_2'(t, 0) = x_2(t), \quad \theta'(t, 0) = \theta(t) \\ \omega'(t, 0) = \omega(t), \quad v'(t, 0) = v(t) \end{cases}$$

and its solution is:

$$\begin{cases} x_1'(t, \tau) = e^\tau x_1(t), \\ x_2'(t, \tau) = e^\tau x_2(t), \\ \theta'(t, \tau) = \theta(t), \\ \omega'(t, \tau) = \omega(t), \\ v'(t, \tau) = e^\tau v(t), \end{cases} \quad (31)$$

which is a scale transformation with factor e^τ . Note that, in accordance with our intuition, this transformation regards all the metric quantities (x_1, x_2, v) .

Let us consider the third generator of $\mathcal{O}^\perp$ in (18). We obtain

$$\xi_1^0 = \frac{\partial \mathcal{L}_{g^0} \tilde{h}_1}{\partial x} \xi = -1, \quad \xi_1^1 = \frac{\partial \mathcal{L}_{g^1} \tilde{h}_1}{\partial x} \xi = 0.$$

From (26) we obtain that the corresponding ${}^u\chi$:

$${}^u\chi_1 = -v_1^1 \xi_1^0 = \frac{\rho}{\sin(\theta - \phi)}, \quad {}^u\chi_2 = 0,$$

where we used the expression of v_1^1 in (12). In this case, we cannot determine the expression of the corresponding indistinguishable v' as the condition in (29) is not met. We can only determine the corresponding indistinguishable v' , which differ from the true v , by an infinitesimal transformation. They are:

$$v' = v + \epsilon {}^u\chi_1 = v + \epsilon \frac{\rho}{\sin(\theta - \phi)}.$$

2.9. Minimal external information

We conclude this section by discussing a fundamental issue that arises every time we are in the presence of unobservability and/or unidentifiability. We wish to answer to the following practical question: *What is the minimal external information (external to the available measurements of the system outputs) required to make observable the state and identifiable all the model parameters?*

In particular, we would like an automatic answer, which holds for any system, as in the case of the observability and identifiability analysis that are obtained by simply running Algorithm 1 and 4.

Unfortunately, we do not have a general answer yet (this will be the matter of a future investigation). For the moment, we have

a preliminary result that provides an answer to our question when the state is characterized by a single symmetry (i.e., the orthogonal distribution $\mathcal{O}^\perp$ has a single generator). The key is that, in the presence of a single symmetry, we are able to determine *all* the indistinguishable states by solving the differential equations in (15) (or even to determine *all* the indistinguishable states and unknown inputs by solving (30)).

Let us denote the components of the true state at time t by $x_1(t), x_2(t), \dots, x_n(t)$. As the system is characterized by a single symmetry, all the indistinguishable states are:

$$\begin{cases} x_1'(t, \tau) = s_1(x_1(t), x_2(t), \dots, x_n(t), \tau) \\ x_2'(t, \tau) = s_2(x_1(t), x_2(t), \dots, x_n(t), \tau) \\ \dots \\ x_n'(t, \tau) = s_n(x_1(t), x_2(t), \dots, x_n(t), \tau) \end{cases} \quad (32)$$

where $s_1, \dots, s_n$ are specific functions determined by solving the differential equations in (15) (or (30), in which case the system in (32) includes the components of the state and the unknown inputs).

We know that, by using all the available measurements, we cannot determine $x_1(t), x_2(t), \dots, x_n(t)$, but we can determine $x_1'(t, \tau), x_2'(t, \tau), \dots, x_n'(t, \tau)$. The latter are related to the former by the equation system in (32). In particular, this system can be regarded as a system of n equations in the $n+1$ unknowns, which are the n components of the true state and τ . Note that we have such a system of equations for any t in our time interval. On the other hand, while the components of the state are in general time dependent, τ is independent of t . Let us suppose that we add a measurement, at a single time t^* , on one of the unobservable components of the true state. Without loss of generality, let us assume that this is the last component of the state, (i.e., we are measuring $x_n(t^*)$). The fact that x_n is unobservable, means that the function s_n depends on τ . Now we consider our equation system in (32) at $t = t^*$. This system consists of n equations in n unknowns, which are the first $n-1$ components of the true state at time t^* , and τ . In general, this allows us to obtain the n unknowns, and in particular, to obtain τ . But once τ has been determined, our equation system in (32) for any $t \neq t^*$ consists of n equations in n unknowns, which are the n components of the true state at time t . In general, this allows us to obtain the n unknowns, i.e., to obtain the components of the state at any time t .

In the next sections, we will see the application of the above property to the viral systems investigated. In particular, it will allow us to determine which measurements must be added to the knowledge of the outputs to obtain observability and identifiability.

3. Observability and identifiability properties of Variant 1

We study the observability and identifiability properties of Variant 1, which is characterized by the dynamics given in (1), by the constant and time varying parameters specified in the second and third column of Table 1, and the outputs specified in the fourth column of Table 1.

3.1. Observability analysis

In Appendix A.1 we provide the computational details for the implementation of Algorithm 1. We summarize the outputs that are obtained:

- The system $\mathcal{E}$ returned by the algorithm coincides with the original system.
- The observability codistribution is the codistribution $\mathcal{O}$ given in (A.5) (and its orthogonal distribution $\mathcal{O}^\perp$ is given in (A.6)).
- The unknown input degree of reconstructability is $m = 1$ and it coincides with m_w (system canonic with respect to its unknown input).

It is immediate to check that T_U and T_I are not observable (it suffices to show that their gradients, i.e., the covectors $[1, 0, 0, 0, 0, 0, 0]$, and $[0, 1, 0, 0, 0, 0, 0]$, are not orthogonal to $\mathcal{O}^\perp$ and, consequently, they do not belong to $\mathcal{O}$).

This result contradicts the result available in the scientific literature (e.g., see Section 6.2 of Miao et al., 2011). The reader is addressed to Section 7 and to Martinelli (2023a) for a detailed discussion about this discrepancy.

3.2. Identifiability analysis

We run Algorithm 4 to obtain the identifiability of all the parameters, constant (λ , ρ , δ , N , c) and time varying ($\eta(t)$). In Appendix A.2 we provide the computational details of this implementation. The result is that only the constant parameters λ , ρ , and c are locally identifiable. The remaining two constant parameters δ and N , and the time varying parameter $\eta(t)$ are unidentifiable, even locally. This result contradicts the result available in the scientific literature (e.g., see Section 6.2 of Miao et al. (2011)). The reader is addressed to Section 7 and to (Martinelli, 2023a) for a detailed discussion about this discrepancy.

3.3. Indistinguishable states and parameters

We use the results given in Sections 2.6 and 2.7 to determine infinitely many values of the state and the parameters, which are indistinguishable from their true values. The details of this computation are available in Appendix A.3. From them, we obtain the following fundamental (new) result that contradicts the current literature. For any choice of τ , the following state and parameters

$$\begin{cases} T'_U(t, \tau) = T_U(t) + T_I(t) - \frac{T_I(t)}{\rho} (\delta e^{-\rho\tau} + \rho - \delta) \\ T'_I(t, \tau) = \frac{T_I(t)}{\rho} (\delta e^{-\rho\tau} + \rho - \delta) \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda \\ \rho'(\tau) = \rho \\ \delta'(\tau) = \delta\rho / ((\rho - \delta)e^{\rho\tau} + \delta) \\ N'(\tau) = Ne^{\rho\tau} \\ c'(\tau) = c \\ \eta'(t, \tau) = \frac{\eta(t)T_U(t)V(t)\rho e^{\rho\tau} + [T_I(t)\delta^2 - T_I(t)\delta\rho - \eta(t)T_U(t)V(t)\delta][e^{\rho\tau} - 1]}{V(t)[T_I(t)\delta + T_U(t)\rho]e^{\rho\tau} - V(t)T_I(t)\delta} \end{cases} \quad (33)$$

are indistinguishable from the true state and the true parameters. In Appendix A.4 we provide an unequivocal and independent proof of their indistinguishability. The model is unobservable and unidentifiable, in contrast with the scientific literature (e.g., as it was obtained in Section 6.2 of Miao et al., 2011).

In the rest of this section, we provide the above indistinguishable states and parameters by using the same data set adopted in Mhawej et al. (2010). The data set is characterized as follows. The time interval is $I = [0, 325]$ (in days). In addition, $T_U(0) = 1000$ CD4/mm³, $T_I(0) = 1$ CD4/mm³, $V(0) = 50$ copies/ml, $\lambda = 9$ mm⁻³ d⁻¹, $\rho = 0.009$ d⁻¹, $\delta = 0.3$ d⁻¹, $N = 250$ mm³ ml⁻¹, and $c = 0.6$ d⁻¹. Finally, we set the time varying parameter $\eta(t)$ as follows:

$$\eta(t) = \eta_0 [0.1 + 0.9 \cos(\alpha_\eta t)], \quad (34)$$

with $\eta_0 = 0.0000041$ ml d⁻¹ and $\alpha_\eta = \frac{\arccos(4/9)}{325}$ d⁻¹ $\cong 0.0034$ d⁻¹. This profile was set with the following criterion. Its time evolution is sinusoidal (e.g., see the electronic supplementary material of Villaverde et al., 2019), its initial value is η_0 , and its final value ($t = 325$ d) is half (from Table 1 in Mhawej et al., 2010 we can see that the ratio between the initial and the final value is around 2).

We provide indistinguishable states and indistinguishable unknown inputs for several values of the parameter $\tau \in \mathcal{U}$, where $\mathcal{U}$ is an interval

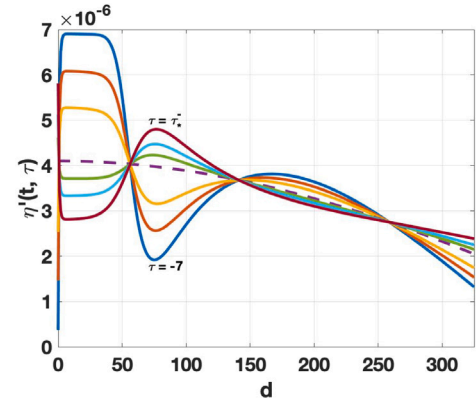


Fig. 2. Several indistinguishable profiles for the time varying parameter $\eta'(t, \tau)$ for several values of the parameter τ ranging from -7 up to $\tau_* \cong 3.3844$. The dashed purple line is the profile given by (34), i.e., $\eta'(t, \tau)$ for $\tau = 0$.

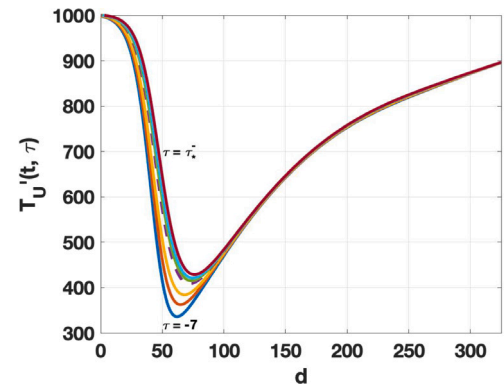


Fig. 3. The profiles $T'_U(t, \tau)$ for several values of the parameter τ ranging from -7 up to $\tau_* \cong 3.3844$. The dashed purple line is the profile for $\tau = 0$.

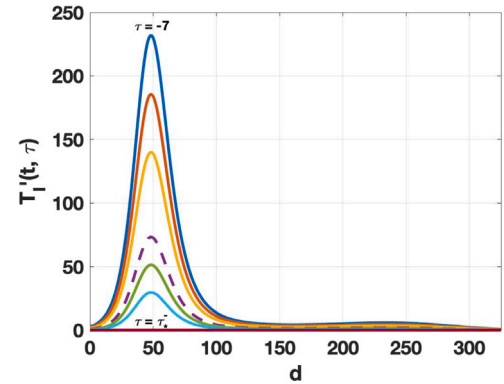


Fig. 4. The profiles $T'_I(t, \tau)$ for the same settings specified in Fig. 3.

of $\mathbb{R}$ that includes $\tau = 0$ and such that, for any $\tau \in \mathcal{U}$, the quantities in (33) have physical meaning, i.e., they are positive at any time.

From the expression of $\delta'(\tau)$ in (33) and the values of our data set, we obtain that $\lim_{\tau \rightarrow \tau_*^-} \delta'(\tau) = +\infty$, $\lim_{\tau \rightarrow \tau_*^+} \delta'(\tau) = -\infty$, with $\tau_* := \frac{\log(\frac{\delta}{\delta-\rho})}{\rho} \cong 3.3844$. In addition, we found that, in the interval $\mathcal{U} := [-7, \tau_*)$ all the quantities $\delta'(\tau)$, $N'(\tau)$, $\eta'(t, \tau)$, $T'_U(t, \tau)$, and $T'_I(t, \tau)$ take positive values (the last three on the entire time interval I). Hence, we are allowed to use any $\tau \in \mathcal{U}$.

Fig. 2 displays the profiles of $\eta'(t, \tau)$. The profile for $\tau = 0$ (purple dashed line) is precisely the one set in (34). By varying τ we obtain

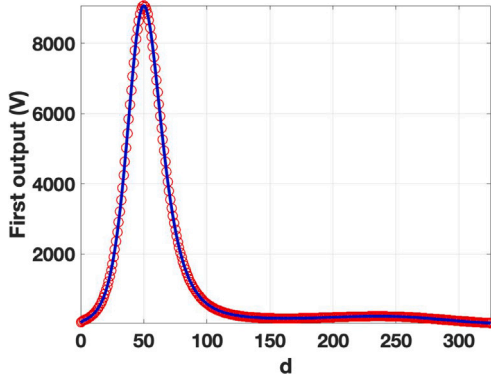


Fig. 5. The first output, $y_1 = V$. The blue solid line displays the output obtained with the true state while red circles display the outputs obtained by generating the indistinguishable state with $\tau = 3.38$ as explained in the text.

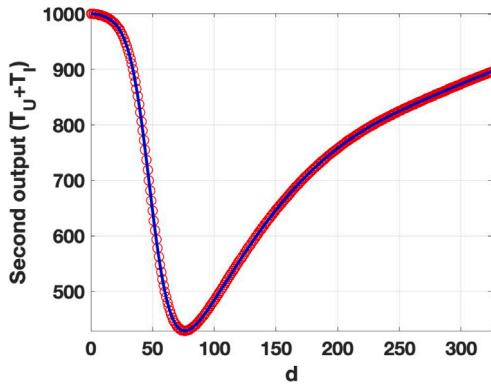


Fig. 6. The second output, $y_2 = T_U + T_I$, with the same settings as in Fig. 5.

a significant change of the profile meaning that this parameter is “strongly” unidentifiable. In addition, by varying τ , the two constant parameters that are not identifiable (i.e., δ and N) significantly change.

Figs. 3 and 4 display the profiles of $T'_U(t, \tau)$ and $T'_I(t, \tau)$. They also change significantly by varying τ meaning that also the first two state components are “strongly” unobservable.

Finally, Figs. 5 and 6 display the two outputs, $y_1 = V$ (Fig. 5) and $y_2 = T_U + T_I$ (Fig. 6). The blue solid line displays the outputs obtained with the true state. The true state at time t is obtained by directly integrating the differential equations in (1) with the constant parameters equal to the true values (i.e., $\lambda = 9 \text{ mm}^{-3} \text{ d}^{-1}$, $\rho = 0.009 \text{ d}^{-1}$, $\delta = 0.3 \text{ d}^{-1}$, $N = 250 \text{ mm}^3 \text{ ml}^{-1}$, and $c = 0.6 \text{ d}^{-1}$), the time varying parameter $\eta(t)$ set as in (34), and the initial conditions set to the true values (i.e., $T_U(0) = 1000 \text{ CD4/ mm}^3$, $T_I(0) = 1 \text{ CD4/ mm}^3$, $V(0) = 50 \text{ copies/ ml}$). Red circles display the outputs obtained by generating indistinguishable states as follows. The state at time t is obtained by directly integrating the differential equations in (1) with the constant parameters equal to the values in (33) for $\tau = 3.38$ (i.e., $\lambda' = \lambda = 9 \text{ mm}^{-3} \text{ d}^{-1}$, $\rho' = \rho = 0.009 \text{ d}^{-1}$, $\delta' = \delta \rho / ((\rho - \delta)e^{\rho\tau} + \delta) = 229.55 \text{ d}^{-1}$, $N' = N e^{\rho\tau} = 257.72 \text{ mm}^3 \text{ ml}^{-1}$, and $c' = c = 0.6 \text{ d}^{-1}$), the time varying parameter $\eta(t)$ set as in the last equation of (33) with $\tau = 3.38$, and the initial conditions equal to the values obtained by using the first three equations in (33) at $t = 0$ and with $\tau = 3.38$ (i.e., $T_U(0)' = T_U(0) + T_I(0) - \frac{T_I(0)}{\rho} (\delta e^{-\rho\tau} + \rho - \delta) = 1001 \text{ CD4/ mm}^3$, $T_I(0)' = \frac{T_I(0)}{\rho} (\delta e^{-\rho\tau} + \rho - \delta) = 0.0013 \text{ CD4/ mm}^3$, $V'(0) = V(0) = 50 \text{ copies/ ml}$). From the profiles displayed in Figs. 5 and 6 it is possible to conclude that the outputs obtained with the true state coincide with the outputs obtained with the states built by using the continuous symmetry of the system, and this directly and unequivocally proves their indistinguishability. Note

that we obtained the same result by using other values of τ and also by using different data sets.

3.4. Minimal external information

Our analysis showed that only three parameters are identifiable. They are λ , ρ , and c . The remaining constant parameters, δ , and N , and the time varying parameter, $\eta(t)$, cannot be identified. It is possible to prove that, by also including the output $y_3 = T_U$ (or $y_3 = T_I$), the state becomes observable and all the parameters identifiable. This can be proved by repeating the analysis for the new system characterized by this further output. On the other hand, from our analysis, it is immediate to obtain a more interesting result.

The system is characterized by a single symmetry. As a result, we can apply the general procedure described in Section 2.9 and we can conclude that it suffices to have T_U (or T_I) at a single time $t^* \in I$ (and not necessarily on the entire time interval) to make possible the identification of all the parameters (and the observability of the state). Eq. (32) becomes, in this case, (33). In order to determine the true state and the true parameters, we must determine the value of τ . This can be determined by knowing T_U (or equivalently $T_I = y_2 - T_U$) at a single time. First of all, Eq. (33) tells us that we have the true V , λ , ρ , and c . Now let us suppose that we know T_I at a single time. From the second equation in (33) we obtain the quantity $\delta e^{-\rho\tau} - \delta = \frac{T'_I(t^*, \tau) \rho}{T_I(t^*)} - \rho$, which is an equation in the two unknowns δ and τ . On the other hand, the sixth equation in (33) is a further equation in the same two unknowns. From them, we obtain both δ and τ . Finally, once τ has been determined, we determine the remaining quantities at any $t \in I$.

3.5. Characterization by an observable state and identifiable parameters

We want to describe the system by only using an observable state and parameters which are identifiable. This forces us to redefine the state and the unidentifiable parameters (constant and time varying). The original state has dimension 8 and the observability codistribution 7. This is the reason why the orthogonal distribution in (A.6) has dimension 1 ($8 = 7 + 1$). A possible observable state consists precisely of a set of generators of the observability codistribution. Algorithm 1 automatically returns a set of generators (see Eq. (A.5)). On the other hand, by redefining the state in this manner, the determination of the expression of its dynamics in terms of the solely components of this state is definitely prohibitive (although possible). To simplify this task, first of all, we try to determine 7 independent and simple functions, which are observable and that generate the observability codistribution. By using the expression of the above generators and the expression of the orthogonal distribution in (A.6), by using our inventiveness, we were able to select the following observable functions:

$$\Psi_1 := V, \Psi_2 := T_U + T_I, \Psi_3 := N\delta T_I, \quad (35)$$

$$\mu := \frac{\delta - \rho}{N\delta}, \lambda, c, \rho.$$

By a direct calculation it is possible to verify that their gradients generate $\mathcal{O}$, i.e., $\mathcal{O} = \text{span}\{\nabla\Psi_1, \nabla\Psi_2, \nabla\Psi_3, \nabla\mu, \nabla\lambda, \nabla c, \nabla\rho\}$. As their expression is much easier than the expression of the generators automatically selected by Algorithm 1, we have the possibility to describe our system by using them. After some trials, by using (1), we were able to obtain the following dynamics:

$$\begin{cases} \dot{\Psi}_1 &= \Psi_3 - c\Psi_1 \\ \dot{\Psi}_2 &= \lambda - \rho\Psi_2 - \Psi_3 \mu \\ \dot{\Psi}_3 &= \tilde{\eta} \\ \dot{\mu} &= 0 \\ \dot{\lambda} &= 0 \\ \dot{c} &= 0 \\ \dot{\rho} &= 0 \\ y &= [\Psi_1, \Psi_2] \end{cases} \quad (36)$$

The correctness of the above expression can be easily verified by using in (36) the expressions of Ψ_1 , Ψ_2 , Ψ_3 , and μ given in (35) and by using Eq. (1). Note that, in order to achieve this description, we had to redefine the unknown input ($\tilde{\eta} := N\delta T_U \Psi_1 \eta - \delta \Psi_3$). By using the original unknown input η , we would obtain $\dot{\Psi}_3 = N\delta T_U \Psi_1 \eta - \delta \Psi_3$. This equation cannot be expressed only in terms of the components of the new observable state. It needs the product $N\delta T_U$ and δ , which cannot be expressed in terms of the selected observable functions because both $N\delta T_U$ and δ are unobservable.

4. Observability and identifiability properties of Variant 2

We study the observability and identifiability properties of Variant 2, which is characterized by the dynamics given in (1), by the constant and time varying parameters specified in the second and third column of Table 1, and the outputs specified in the fourth column of Table 1.

4.1. Observability analysis

In Appendix B.1 we provide the computational details for the implementation of Algorithm 1. We summarize the outputs that are obtained:

- The system $\mathcal{E}$ returned by the algorithm coincides with the original system.
- The observability codistribution is the codistribution $\mathcal{O}$ given in (B.5) (and its orthogonal distribution $\mathcal{O}^\perp$ is given in (B.6)).
- The unknown input degree of reconstructability is $m = 1$ and it coincides with m_w (system canonic with respect to its unknown input).

It is immediate to check that T_U , T_I , and V are observable (it suffices to show that their gradients, i.e., the covectors $[1, 0, 0, 0, 0, 0, 0]$, $[0, 1, 0, 0, 0, 0, 0]$, and $[0, 0, 1, 0, 0, 0, 0]$, are orthogonal to $\mathcal{O}^\perp$ and, consequently, they belong to $\mathcal{O}$).

4.2. Identifiability analysis

We use Algorithm 4 to obtain the identifiability of all the parameters, constant (λ , ρ , η , δ , c) and time varying ($N(t)$). In Appendix B.2 we provide the computational details of this implementation. The result is that all the constant parameters are locally identifiable with the exception of c . In addition, the time varying parameter $N(t)$ is unidentifiable, even locally.

4.3. Indistinguishable states and parameters

We use the results given in Sections 2.6 and 2.7 to determine infinitely many values of the state and the parameters, which are indistinguishable from their true values. The details of this computation are available in Appendix B.3. From them, we obtain the following new result. For any choice of τ , the parameters c' and N' given in (37) are indistinguishable from c and N .

$$\left\{ \begin{array}{l} T'_U(t, \tau) = T_U(t) \\ T'_I(t, \tau) = T_I(t) \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda \\ \rho'(\tau) = \rho \\ \eta'(\tau) = \eta \\ \delta'(\tau) = \delta \\ c'(\tau) = c + \tau \\ N'(t, \tau) = N(t) + \frac{V}{T_I \delta} \tau \end{array} \right. \quad (37)$$

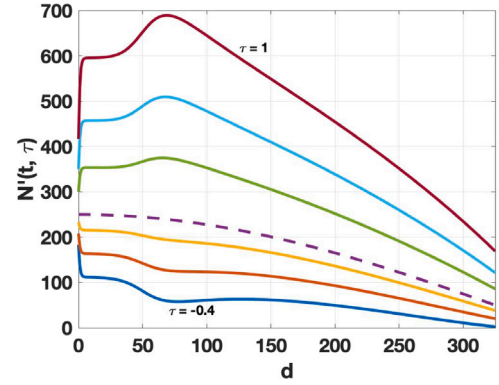


Fig. 7. Several indistinguishable profiles for the time varying parameter $N'(t, \tau)$ for several values of the parameter τ ranging from -0.4 up to 1 . The dashed purple line is the profile given by (38), i.e., $N'(t, \tau)$ for $\tau = 0$.

In the rest of this section, we provide the above indistinguishable states and parameters by using the same data set adopted in Section 3.3. In this case, $\eta = \eta_0$ and N is set as follows:

$$N(t) = N_0 [0.1 + 0.9 \cos(\alpha_N t)], \quad (38)$$

with $N_0 = 250 \text{ mm}^3 \text{ ml}^{-1}$ and $\alpha_N = \frac{\arccos(1/9)}{325} \text{ d}^{-1} \cong 0.0045 \text{ d}^{-1}$. This profile provides the initial value N_0 , and the final value (i.e., at $t = 325 \text{ d}$) one fifth of the initial value.

We provide indistinguishable unknown inputs for several values of the parameter $\tau \in \mathcal{U}$, where $\mathcal{U}$ is an interval of $\mathbb{R}$ that includes $\tau = 0$ and such that, for any $\tau \in \mathcal{U}$, the quantities in (37) take positive values. We found that this is certainly the case by setting $\mathcal{U} := [-0.4, \infty)$. Hence, we are allowed to use any $\tau \in \mathcal{U}$.

Fig. 7 displays the profiles of $N'(t, \tau)$ for $\tau \in [-0.4, 1] \subset \mathcal{U}$. The profile for $\tau = 0$ (purple dashed line) is precisely the one set in (38). Even by varying $\tau \in [-0.4, 1]$, we obtain a significant change of the profile meaning that this parameter is “strongly” unidentifiable. In addition, by varying τ , the constant parameter that is not identifiable (i.e., c) significantly changes.

We do not provide the profiles of T'_U , T'_I , and V' , as all these components of the state are observable (and consequently independent of τ). For brevity we do not even provide the outputs obtained with the true state and with the indistinguishable states, as we did at the end of Section 3.3 for the first variant. We only mention that, also in this case, they perfectly coincide, and the outputs are very similar to the ones displayed in Figs. 5 and 6.

4.4. Minimal external information

Also in this case the system is characterized by a single symmetry and we can use the general procedure in Section 2.9. In this case, the only component of the state in (B.1) which is unobservable is c . Hence, by a single measurement of c we break the symmetry and we gain observability. Specifically, from the expression of $c'(\tau)$ in (37), we trivially have $\tau = c'(\tau) - c$.

On the other hand, for this system, we obtained the solution given in (37), which also includes the expression of the time varying parameter. We can extend the general procedure given in Section 2.9 to also the time varying parameters, i.e., we can use (37) as a specific case of the system in (32). We conclude that, another possibility to obtain observability and identifiability consists of a single measurement at a single time $t^* \in I$ of the time varying parameter $N(t)$. Indeed, from the expression of $N'(t, \tau)$ in (37), we trivially have $\tau = [N'(t^*, \tau) - N(t^*)] \frac{T_I(t^*) \delta}{V(t^*)}$. Finally, once we have τ , we have $N(t)$ at any time, as $N(t) = N'(t, \tau) - \frac{V}{T_I \delta} \tau$.

4.5. Characterization by an observable state and identifiable parameters

We want to describe the system by only using an observable state and parameters which are identifiable. We proceed as in Section 3.5. We first determine a set of 7 simple observable functions that generate the entire observability codistribution. In this case this task is trivial as we know that all the components of the state are observable, with the exception of the last one (c). Therefore, $\mathcal{O} = \text{span}\{\nabla T_U, \nabla T_I, \nabla V, \nabla \lambda, \nabla \rho, \nabla \eta, \nabla \delta\}$. The description in terms of the state which consists of the above generators is trivially:

$$\begin{cases} \dot{T}_U &= \lambda - \rho T_U - \eta T_U V \\ \dot{T}_I &= \eta T_U V - \delta T_I \\ \dot{V} &= \tilde{N} \\ y &= [V, T_U + T_I], \end{cases} \quad (39)$$

Note that we redefine the unknown input ($\tilde{N} := N\delta T_I - cV$). By using the original unknown input N , we would obtain $\dot{V} = N\delta T_I - cV$. This equation cannot be expressed only in terms of the components of the new observable state, as c is unidentifiable.

5. Observability and identifiability properties of Variant 3

We study the observability and identifiability properties of Variant 3, which is characterized by the dynamics given in (1), by the constant and time varying parameters specified in the second and third column of Table 1, and the outputs specified in the fourth column of Table 1.

5.1. Observability analysis

In Appendix C.1 we provide the computational details for the implementation of Algorithm 1. We summarize the outputs that are obtained:

- The system $\mathcal{E}$ returned by the algorithm coincides with the original system.
- The observability codistribution is the codistribution $\mathcal{O}$ given in (C.5) (and its orthogonal distribution $\mathcal{O}^\perp$ is given in (C.6)).
- The unknown input degree of reconstructability is $m = 2$ and it coincides with m_w (system canonic with respect to its unknown inputs).

T_U , T_I , and V are observable as their gradient is orthogonal to the distribution $\mathcal{O}^\perp$ given in (C.6)). This is not a surprise because they are the system outputs.

5.2. Identifiability analysis

We use Algorithm 4 to obtain the identifiability of all the parameters, constant (λ , ρ , δ , c) and time varying ($\eta(t)$, $N(t)$). In Appendix C.2 we provide the computational details of this implementation. The result is that all the constant parameters are locally identifiable with the exception of c . In addition, the time varying parameter $N(t)$ is unidentifiable, even locally. In contrast, the time varying parameter $\eta(t)$ is locally identifiable.

5.3. Indistinguishable states and parameters

We use the results given in Sections 2.6 and 2.7 to determine infinitely many values of the state and the parameters, which are indistinguishable from their true values. The details of this computation are available in Appendix C.3. From them, we have obtained the

following new result. For any choice of τ , the parameters c' and N' given in (40) are indistinguishable from c and N .

$$\begin{cases} T'_U(t, \tau) &= T_U(t) \\ T'_I(t, \tau) &= T_I(t) \\ V'(t, \tau) &= V(t) \\ \lambda'(\tau) &= \lambda \\ \rho'(\tau) &= \rho \\ \delta'(\tau) &= \delta \\ c'(\tau) &= c + \tau \\ \eta'(t, \tau) &= \eta(t) \\ N'(t, \tau) &= N(t) + \frac{V}{T_I \delta} \tau \end{cases} \quad (40)$$

The profiles of $N'(t, \tau)$ are exactly the same obtained in Section 4.3, which are displayed in Fig. 7 for our data set. For brevity, we do not provide the outputs obtained with the true state and with the indistinguishable states, as we did at the end of Section 3.3 for the first variant. We only mention that, also in this case, they perfectly coincide, and the outputs are very similar to the ones displayed in Figs. 5 and 6.

5.4. Minimal external information

We have exactly the same properties provided in Section 4.4.

5.5. Characterization by an observable state and identifiable parameters

We have the same description provided in Section 4.5. In particular, the observable state is the same and also the dynamics (first three equations in (39)). For the outputs, instead of the last equation in (39), we have: $y = [V, T_U, T_I]$.

6. Observability and identifiability properties of Variant 4

We study the observability and identifiability properties of Variant 4, which is characterized by the dynamics given in (1), by the constant and time varying parameters specified in the second and third column of Table 1, and the outputs specified in the fourth column of Table 1.

6.1. Observability analysis

In Appendix D.1 we provide the computational details for the implementation of Algorithm 1. We summarize the outputs that are obtained:

- The system $\mathcal{E}$ returned by the algorithm coincides with the original system.
- The observability codistribution is the codistribution $\mathcal{O}$ given in (D.4) (and its orthogonal distribution $\mathcal{O}^\perp$ is given in (D.5)).
- The unknown input degree of reconstructability is $m = 2$ and it coincides with m_w (system canonic with respect to its unknown inputs).

It is immediate to check that T_U and T_I are not observable (it suffices to show that their gradients, i.e., the covectors $[1, 0, 0, 0, 0, 0, 0]$, $[0, 1, 0, 0, 0, 0, 0]$ are not orthogonal to $\mathcal{O}^\perp$ (they are not orthogonal to the first three generators of $\mathcal{O}^\perp$) and, consequently, they cannot belong to $\mathcal{O}$).

6.2. Identifiability analysis

We use Algorithm 4 to obtain the identifiability of all the parameters, constant (λ , ρ , δ , c) and time varying ($N(t)$, $\eta(t)$). In Appendix D.2 we provide the computational details of this implementation. The result is that all these parameters are unidentifiable, even locally.

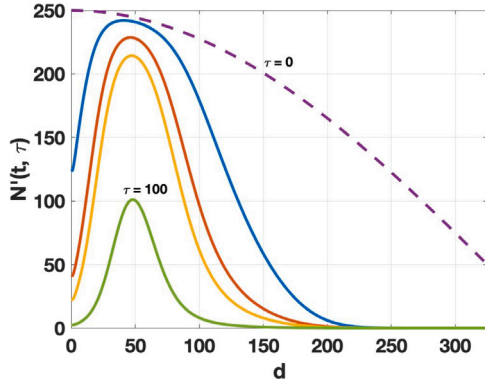


Fig. 8. The profiles $N'(t, \tau)$ for several values of the parameter τ ranging from 0 up to 100. The dashed purple line is the profile for $\tau = 0$.

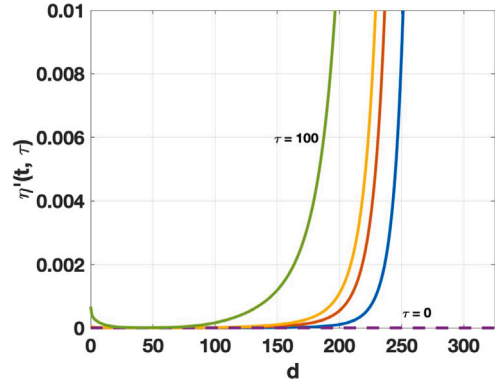


Fig. 9. The profiles $\eta'(t, \tau)$ for the same settings specified in Fig. 8.

6.3. Indistinguishable states and parameters

We use the results given in Sections 2.6 and 2.7 to determine infinitely many values of the state and the parameters, which are indistinguishable from their true values. The details of this computation are available in Appendix D.3. With respect to the previous variants, which are characterized by a single continuous symmetry, in this variant the orthogonal distribution $\mathcal{O}^\perp$ has four generators. As a result, we can determine four distinct sets of indistinguishable states and parameters.

6.3.1. First set

In Appendix D.3.1, we determine the indistinguishable states and parameters by using the first generator of $\mathcal{O}^\perp$ in (D.5). They are given in (41).

$$\begin{cases} T'_U(t, \tau) = T_U(t) - \tau \\ T'_I(t, \tau) = T_I(t) + \tau \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda + (\delta - \rho)\tau \\ \rho'(\tau) = \rho \\ \delta'(\tau) = \delta \\ c'(\tau) = c \\ N'(t, \tau) = N(t) \frac{T_I(t)}{T_I(t) + \tau} \\ \eta'(t, \tau) = \frac{\delta\tau}{V(t)[T_U(t) - \tau]} + \eta(t) \frac{T_U(t)}{T_U(t) - \tau} \end{cases} \quad (41)$$

We explicitly provide them for the same data set adopted in Section 3.3. The only difference is that now both the parameters N and η are time varying. Their profile are set by using (38) and (34), respectively.

We found that, by setting $\mathcal{U} := [0, 100]$, all the solutions in (41) are positive. Hence, we are allowed to use any $\tau \in \mathcal{U}$.

Figs. 8 and 9 display the profiles of $N'(t, \tau)$ and $\eta'(t, \tau)$. The profiles for $\tau = 0$ (purple dashed line) are precisely the ones set in (38), and in (34), respectively. By varying τ we obtain a significant change of the profiles meaning that these parameters are “strongly” unidentifiable. In addition, by varying τ , the constant parameter that is not identifiable (i.e., λ) significantly changes. The profiles of T'_U, T'_I are simple shifts of the true profiles ($-\tau$ and $+\tau$, respectively).

6.3.2. Second set

In Appendix D.3.2, we determine the indistinguishable states and parameters by using the second generator of $\mathcal{O}^\perp$ in (D.5). They are given

in (42).

$$\begin{cases} T'_U(t, \tau) = T_U(t)e^\tau \\ T'_I(t, \tau) = T_I(t) + T_U(t)[1 - e^\tau] \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda \\ \rho'(\tau) = \rho e^{-\tau} + \delta(1 - e^{-\tau}) \\ \delta'(\tau) = \delta \\ c'(\tau) = c \\ N'(t, \tau) = N(t) \frac{T_I(t)}{T_I(t) + T_U(t)[1 - e^{-\tau}]} \\ \eta'(t, \tau) = \eta(t) + \frac{\lambda + T_U(t)[\delta - \rho]}{V(t)T_U(t)}(e^{-\tau} - 1) \end{cases} \quad (42)$$

We do not display them. Indeed, for the data set adopted in Section 3.3, the range of variability of τ that guarantees the positivity of these solutions is very small.

6.3.3. Third set

In Appendix D.3.3, we determine the indistinguishable states and parameters by using the third generator of $\mathcal{O}^\perp$ in (D.5). They are given in (43).

$$\begin{cases} T'_U(t, \tau) = T_U(t) + T_I(t)[1 - e^{-\tau}] \\ T'_I(t, \tau) = T_I(t)e^{-\tau} \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda \\ \rho'(\tau) = \rho \\ \delta'(\tau) = \delta e^\tau + \rho(1 - e^\tau) \\ c'(\tau) = c \\ N'(t, \tau) = N(t) \frac{\delta}{\delta + \rho(e^\tau - 1)} \\ \eta'(t, \tau) = \frac{T_I(t)(\delta - \rho)(1 - e^{-\tau}) + \eta(t)V(t)T_U(t)e^{-\tau}}{V(t)T_U(t) + V(t)T_I(t)[1 - e^{-\tau}]} \end{cases} \quad (43)$$

We explicitly provide them for the same data set adopted in Section 3.3. As for the first set of solutions, both the parameters N and η are time varying and, as in that case, they are set by using the two profiles in (38) and (34), respectively.

We found that, by setting $\mathcal{U} := [-0.3, 10]$, all the solutions in (43) are positive. Hence, we are allowed to use any $\tau \in \mathcal{U}$.

Figs. 10 and 11 display the profiles of $N'(t, \tau)$ and $\eta'(t, \tau)$. The profiles for $\tau = 0$ (purple dashed line) are precisely the ones set in (38), and in (34), respectively. By varying τ we obtain a significant change of the profile of $\eta'(t, \tau)$, meaning that this parameter is “strongly” unidentifiable. In addition, by varying τ , the constant parameter that is not identifiable (i.e., δ) significantly changes.

Figs. 12 and 13 display the profiles of $T'_U(t, \tau)$ and $T'_I(t, \tau)$. The latter changes significantly by varying τ meaning that T_I is “strongly” unobservable.

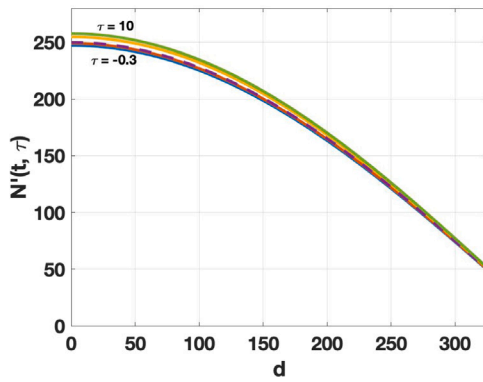


Fig. 10. The profiles $N'(t, \tau)$ for several values of the parameter τ ranging from -0.3 up to 10 . The dashed purple line is the profile for $\tau = 0$.

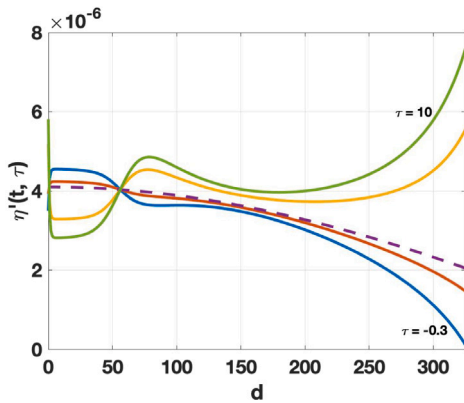


Fig. 11. The profiles $\eta'(t, \tau)$ for the same settings specified in Fig. 10.

6.3.4. Fourth set

In Appendix D.3.4, we determine the indistinguishable states and parameters by using the fourth generator of $\mathcal{O}^\perp$ in (D.5). They are given in (44).

$$\begin{cases} T'_U(t, \tau) = T_U(t) \\ T'_I(t, \tau) = T_I(t) \\ V'(t, \tau) = V(t) \\ \lambda'(\tau) = \lambda \\ \rho'(\tau) = \rho \\ \delta'(\tau) = \delta \\ c'(\tau) = c + \tau \\ N'(t, \tau) = N(t) + \frac{V}{T_I \delta} \tau \\ \eta'(t, \tau) = \eta(t) \end{cases} \quad (44)$$

The profiles of $N'(t, \tau)$ are exactly the same obtained in Section 4.3, which are displayed in Fig. 7 for our data set.

For brevity we do not provide the outputs obtained with the true state and with the indistinguishable states for any of the previous four sets (as we did at the end of Section 3.3 for the first variant). We only mention that, also in this case, they perfectly coincide, and the outputs are very similar to the ones displayed in Figs. 5 and 6.

6.4. Minimal external information

This time, the system has multiple (four) symmetries and we cannot apply the general procedure given in Section 2.9. We obtain an answer

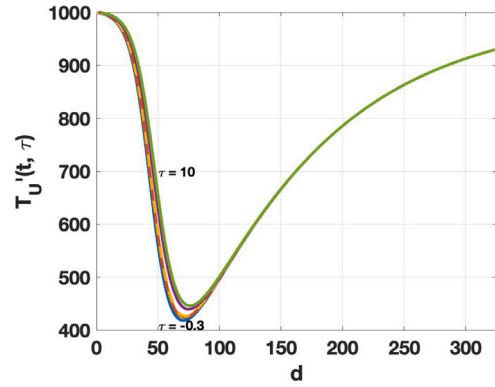


Fig. 12. The profiles $T'_U(t, \tau)$ for the same settings specified in Fig. 10.

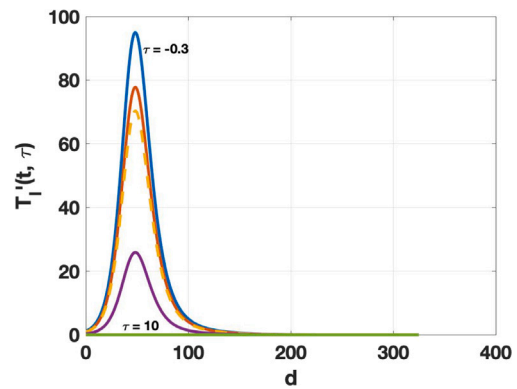


Fig. 13. The profiles $T'_I(t, \tau)$ for the same settings specified in Fig. 10.

to our question by using a heuristic and ad hoc procedure, which exploits the fact that many of the components of the state (given in (C.1)) are constant. The observability analysis in Section 6.1 reveals that, from the measurements, we can only obtain: $T_U + T_I$, V , and $\lambda - T_I \delta - T_U \rho$ (their gradients generate the entire observability codistribution). The minimal information to obtain λ , δ , ρ consists of three measurements of T_U (or T_I), at three distinct times. Once we know λ , δ , ρ , from the knowledge of $T_U + T_I$ and $\lambda - T_I \delta - T_U \rho$ at any time, we obtain T_U and T_I at any time. As a result, with these three measurements all the components of the state (T_U, T_I, V) can be determined at any time. In addition, the first three sets of indistinguishable states and parameters in (41), (42), and (43), disappear. The only indistinguishable states/parameters remain the ones given in (44). These are exactly the same as in (40). As a result, to also make c and $N(t)$ identifiable, we have exactly the same result obtained for Variant 3 (which is also obtained for Variant 2, and discussed in Section 4.4). We need a further measurement of $N(t)$ at a single time or, alternatively, the knowledge of c makes identifiable $N(t)$ (at any time).

6.5. Characterization by an observable state and identifiable parameters

We want to describe the system by only using an observable state and parameters which are identifiable. In this case the observable codistribution has dimension 3 and, the expression of all its generators directly returned by Algorithm 1, is trivial. As a result we adopt them, i.e., we define: $\Psi_1 := V$, $\Psi_2 := T_U + T_I$, $\Psi_3 := \lambda - T_I \delta - T_U \rho$. We obtain:

$$\begin{cases} \dot{\Psi}_1 = \tilde{N} \\ \dot{\Psi}_2 = \Psi_3 \\ \dot{\Psi}_3 = \tilde{\eta} \\ y = [\Psi_1, \Psi_2] \end{cases} \quad (45)$$

Note that, in order to achieve this description, we had to redefine both the unknown inputs: $\tilde{\eta} := -\rho\lambda + \delta^2 T_I + (\rho^2 + \rho\eta V - \delta\eta V)T_U$ and $\tilde{N} := N\delta T_I - cV$.

7. Discussion

In this section we compare our approach (which consists of the execution of Algorithms 1 and 4 together with the procedures to determine indistinguishable states and parameters and to characterize the minimal information), with existing/popular methods. We refer to the HIV models here analysed but we also provide more general remarks (Sections 7.2, 7.3). We compare our results with the ones obtained in the following two studies:

- The analysis in Section 6.2 of Miao et al. (2011).
- The analysis in Section 3.3 of Villaverde et al. (2019).

Both of them studied the identifiability of our first variant in Section 3 and, in both cases, the authors concluded that the system is observable and identifiable. We discuss the origin of the discrepancies with our results on this specific case (Section 7.1) and more general issues (Sections 7.2, 7.3).

7.1. Origin of the discrepancies

First of all, we highlight that the system investigated in Section 3 is unobservable and unidentifiable. We provided the following two unequivocal and independent proofs:

1. (Analytical): The analytical proof given in Appendix A.4, which shows that T'_U , T'_I , V' , η' , λ' , δ' , ρ' , N' , and c' given in (33) cannot be distinguished from T_U , T_I , V , η , λ , δ , ρ , N , and c , respectively.
2. (Numerical): The indistinguishability of the outputs in Figs. 5 and 6 (see also the discussion at the end of Section 3.3).

Hence, the two analyses in Section 6.2 of Miao et al. (2011) and in Section 3.3 of Villaverde et al. (2019) contain errors.

The reader is addressed to Section 3 of Martinelli (2023a) for a detailed study of the errors made by the authors of Miao et al. (2011). Note that Section 6.2 of Miao et al. (2011) contains two distinct errors. The former, which is a simple typo, was recently acknowledged by the authors (see Miao et al., 2023). However, in Miao et al. (2023), the authors still concluded that the system is observable/identifiable. The serious error committed by the authors is highlighted in Section 3.2 of Martinelli (2023a) and lies in an incorrect use of the Inverse Function Theorem. In particular, using this theorem, the authors did not take into account the constraints due to the dynamics of the system, as shown in Section 3.2 of Martinelli (2023a).

Regarding the analysis provided in Villaverde et al. (2019), the error lies in the assumption made in Section 2 of Villaverde et al. (2019) (Assumption 2.5). Let us analyse this assumption in general (and not simply for the specific system). This assumption is inappropriate and does not hold in many cases (e.g., the sinus function does not satisfy this assumption). More importantly, the inappropriateness of this assumption is much more subtle and can make the result of the entire observability analysis spurious. This occurs whenever the system is unobservable but the following property holds: *The knowledge of the time derivative of a given order of the time varying parameter makes the system observable*. By artificially setting this time derivative to zero, Assumption 2.5 makes this time derivative known. The result of the observability analysis is evidently false. This is precisely what happens in our specific case. Indeed, the system is unobservable/unidentifiable, and, by a direct computation, it is possible to show that it becomes observable/identifiable by adding the new output $h_3 = \frac{d^k \eta}{dt^k}$ for a given k , independently of the value of k (which can be as large as we want).

7.2. Inadequacy of existing methods

As we stated in the introduction, all the methods available in the scientific literature cannot provide the observability and/or the identifiability of nonlinear systems in the presence of time varying parameters. As a result, for these systems, it is not surprising to find errors in the literature.

In the presence of time varying parameters, and when the system is unobservable or unidentifiable, all the recursive algorithms based on the state augmentation (e.g., Belo et al., 2010; Martinelli, 2015; Villaverde et al., 2019; Maes et al., 2019; Villaverde et al., 2016) do not converge, automatically. As a result, all these algorithms remain inconclusive. Trying to stop their execution with artificial assumptions can be catastrophic, as we highlighted in Section 7.1.

Similarly, the method adopted in Miao et al. (2011) cannot deal with systems characterized by time varying parameters. To account for the presence of the time varying parameter $\eta(t)$, the authors of Miao et al. (2011) adopted a heuristic and ad hoc procedure. Unfortunately, they achieved an erroneous result, as we highlighted in Section 7.1.¹⁶ We also emphasize that the method adopted in Miao et al. (2011) is limited because can only deal with polynomial nonlinearities.

7.3. Further differences/novelities

We wish to highlight that, in addition to Algorithms 1 and 4, our approach consists of the exploitation of the concept of continuous symmetry to introduce the new procedures described in Section 2 to generate indistinguishable states and parameters (Sections 2.4, 2.6, and 2.7) and to characterize the minimal information to achieve observability and identifiability 2.9. As far as we know, no investigation has addressed these issues to date.

8. Conclusion

This paper provided the observability and identifiability properties of an important ODE model, commonly used to characterize the dynamics of the HIV. We studied four variants of this model. The first variant was already investigated in the past (e.g., Perelson and Nelson, 1999; Perelson, 2002; Miao et al., 2011; Villaverde et al., 2019). We executed Algorithm 1 and 4 to obtain the observability of the state and the identifiability of the model parameters (both constant and time varying). For the first variant, our results are in contrast with the current scientific literature. The interested reader can find a detailed explanation of the error made by the authors of Miao et al. (2011) in Martinelli (2023a). In addition, for all the four variants, we determined the minimal external information (external to the knowledge of the outputs) required to uniquely determine the model parameters and to make observable the state. Finally, for each variant, we concluded by providing a full description of the system only in terms of a state which is observable and a set of parameters which are identifiable. Note that this last step cannot be performed automatically. In addition, this task is not required either to obtain the observability properties or to obtain the identifiability properties. On the other hand, we were able to obtain this description and we strongly believe that it can be useful for further investigations.

We wish also to emphasize that this paper provided a complete implementation of Algorithms 1 and 4. These two systematic procedures can be executed without restrictions on the system investigated. The system was modelled by Eq. (3), and, in Section 2.1, we showed that any system modelled by (2) can be easily converted to a system modelled by (3).

¹⁶ Note that this ad hoc procedure gave a true result, i.e. if all constant parameters are identifiable the time varying parameter $\eta(t)$ is also identifiable.

The theoretical foundation that ensures the general validity of Algorithms 1 and 4 is available in Martinelli (2022).

Algorithms 1 and 4 were used to determine the observability and identifiability properties of the four variants of the HIV model. In addition, for each variant, for the components of the state which are unobservable and for the parameters which are unidentifiable (constant and time varying), we also generated infinitely many values of them which are distinct from the true values but indistinguishable from them. We also directly showed that these indistinguishable values reproduce exactly the same outputs. This is an independent and unequivocal proof of the validity of all the observability and identifiability properties determined in Sections 3–6. The mathematical tool that generates these indistinguishable values consists of a system of first order ordinary differential equations (Eq. (30)). For the first variant the system of differential equations in (30) becomes the system in (A.9). Although this differential equation system can be solved analytically (the solution is given in (33)), it is quite complex. It is perhaps not surprising that, in the scientific literature, no values of the unidentifiable parameters, different from the true ones but indistinguishable from them, were detected, and the model was erroneously classified *observable* and *identifiable*, despite the persistent investigations carried out on this model during the last decades.

Declaration of competing interest

The author declares that no competing financial interests that could have appeared to influence the work reported in this paper.

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Appendix A. Analytical computations for Variant 1

A.1. Implementation of Algorithm 1

We run Algorithm 1 to obtain the observability codistribution. To proceed, we need, first of all, to introduce a state that includes both T_U , T_I , V , and all the constant parameters (i.e., λ , ρ , δ , N , c). We set:

$$x = [T_U, T_I, V, \lambda, \rho, \delta, N, c]^T \quad (\text{A.1})$$

The system is directly a special case of (3). In particular, $m_u = 0$, $m_w = 1$, $p = 2$, $h_1(x) = V$, $h_2(x) = T_U + T_I$,

$$g^0 = \begin{bmatrix} \lambda - \rho T_U \\ -\delta T_I \\ N\delta T_I - cV \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad g^1 = \begin{bmatrix} -T_U V \\ T_U V \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{A.2})$$

Line 2 provides:

$$\Omega = \text{span}\{\nabla h_1, \nabla h_2\} \\ = \text{span}\{[0, 0, 1, 0, 0, 0, 0, 0], [1, 1, 0, 0, 0, 0, 0, 0]\}.$$

We have $\deg_w(\Omega) = 0$ ($< m_w$), as $\mathcal{L}_{g^1} h_1 = \mathcal{L}_{g^1} h_2 = 0$ (and the rank of the unknown input reconstructability matrix from h_1 , h_2 vanishes). The system is not in canonical form with respect to its unknown input. We enter the loop at lines 9–14.

Line 11 provides no functions ($m = 0$). Line 12 provides: $\mu_0^0 = \nu_0^0 = 1$ obtained from (4) (these tensors have a single entry as the indices only

take the value 0). In addition, from (5) we obtain: $\hat{g}^0 = \nu_0^0 g^0 = g^0$. Line 13 computes

$$\Omega = \left\langle \left\{ \begin{bmatrix} g^0 \\ g^1 \end{bmatrix} \right\} \middle| \text{span}\{\nabla h_1, \nabla h_2\} \right\rangle,$$

as $\hat{g}^0 = g^0$. By a direct computation, we obtain:

$$\Omega = \text{span}\{\nabla h_1, \nabla h_2, \nabla \mathcal{L}_{g^0} h_1, \nabla \mathcal{L}_{g^0} h_2\} =$$

$$\text{span}\{\nabla h_1, \nabla h_2, \nabla h_3, \nabla h_4\}$$

where we adopted the following notation:

$$h_3 := \mathcal{L}_{g^0} h_1, \quad h_4 := \mathcal{L}_{g^0} h_2$$

Now, $\deg_w(\Omega) = 1 = m_w$. Hence, the second condition at Line 9 is not met and the loop is interrupted. In addition, as the condition at Line 15 is met, also the main loop is interrupted. The execution of Algorithm 1 continues with Line 45. The operation at this line ($[\tilde{h}_1, \dots, \tilde{h}_{m_w}] = S(\Sigma, \Omega)$) sets

$$\tilde{h}_1 = h_4 = \mathcal{L}_{g^0} h_2 = \lambda - \rho T_U - \delta T_I \quad (\text{A.3})$$

(we would obtain the same final result by setting $\tilde{h}_1 = h_3 = \mathcal{L}_{g^0} h_1$). Then, Line 46 provides:

$$\mu_0^0 = 1, \quad \mu_1^0 = 0, \quad \mu_1^1 = T_I \delta^2 - \rho(\lambda - T_U \rho), \quad \mu_1^1 = -T_U V(\delta - \rho),$$

$$\nu_0^0 = 1, \quad \nu_0^1 = 0, \quad \nu_1^0 = \frac{T_I \delta^2 + T_U \rho^2 - \lambda \rho}{T_U V(\delta - \rho)}, \quad (\text{A.4})$$

$$\nu_1^1 = -\frac{1}{T_U V(\delta - \rho)},$$

obtained from (4). In addition, from (5) we obtain:

$$\hat{g}^0 = \begin{bmatrix} -\frac{\delta(T_I \delta - \lambda + T_U \rho)}{\delta - \rho} \\ \frac{\rho(T_I \delta - \lambda + T_U \rho)}{\delta - \rho} \\ NT_I \delta - Vc \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{g}^1 = \begin{bmatrix} \frac{1}{\delta - \rho} \\ -\frac{1}{\delta - \rho} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Finally, from Equation (20) in Martinelli (2022) we obtain that, when $m_u = 0$, the codistribution $\tilde{\mathcal{O}}$ vanishes. The final step of Algorithm 1 is the execution of Line 47, namely:

$$\mathcal{O} = \langle \hat{g}^0, \hat{g}^1 \mid \text{span}\{\nabla h_1, \nabla h_2, \nabla h_3, \nabla h_4\} \rangle.$$

The algorithm in (1.45) in Isidori (1995) converges at the third step ($\Omega_3 = \Omega_2$), and $\mathcal{O} =$

$$\text{span}\left\{\nabla h_1, \nabla h_2, \nabla h_3, \nabla h_4, \nabla \mathcal{L}_{\hat{g}^0} h_3, \nabla \mathcal{L}_{\hat{g}^1} h_3, \nabla \mathcal{L}_{\hat{g}^0}^2 h_3\right\}. \quad (\text{A.5})$$

Its dimension is 7, which is smaller than the dimension of the state in (A.1). Hence, this state is not observable. We wish to determine the observability properties of T_U , T_I , and V , i.e., the first three components of the state in (A.1). We compute the distribution orthogonal to the observability codistribution in (A.5). We obtain:

$$\mathcal{O}^\perp = \text{span}\left\{\begin{bmatrix} T_I \delta \\ -T_I \delta \\ 0 \\ 0 \\ 0 \\ 0 \\ \delta(\delta - \rho) \\ N\rho \\ 0 \end{bmatrix}\right\}. \quad (\text{A.6})$$

A.2. Implementation of Algorithm 4

The first loop (Lines 2–8) checks the identifiability of the five constant parameters. It is immediate to verify that two of them (δ and N) are unidentifiable (it suffices to show that their gradient, i.e., the covectors $[0, 0, 0, 0, 1, 0, 0]$ and $[0, 0, 0, 0, 0, 1, 0]$, are not orthogonal to $\mathcal{O}^\perp$ and, consequently, they do not belong to $\mathcal{O}$).

The condition at Line 9 is not satisfied (the system is canonic and $m = m_w = 1$). The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (A.4). Line 14 provides the distribution given in (A.6).

The loop at Lines 15–18 consists of the single value $i_s = 1$. For this value, i.e., for the generator of $(\mathcal{O})^\perp$ in (A.6), we compute the coefficients ξ_i^α , for $\alpha = 0, 1$, and $i = 1$, by using (19), (A.2), (A.3), and (A.6). We have:

$$\mathcal{L}_{g^0} \tilde{h}_1 = T_I \delta^2 - \rho(\lambda - T_U \rho), \quad \mathcal{L}_{g^1} \tilde{h}_1 = -T_U V(\delta - \rho),$$

and

$$\Xi_1^{0,1} = \xi_1^0 = T_I \delta(\delta - \rho)^2, \quad \Xi_1^{1,1} = \xi_1^1 = -V \delta(T_I + T_U)(\delta - \rho). \quad (\text{A.7})$$

The quantity in (20) for $\alpha = 0$ becomes:

$$C_1^{0,1} = \sum_{i=1}^m v_i^i \Xi_i^{0,1} = v_1^1 \Xi_1^{0,1} = -\frac{T_I \delta(\delta - \rho)}{T_U V} \neq 0$$

where we used the expression of v_1^1 in (A.4).

As a result, the condition at Line 20 is not honoured, meaning that the time varying parameter $\eta(t)$ is not locally identifiable.

A.3. Application of the method in Sections 2.6 and 2.7

First of all, the system is canonic with respect to its UI. As a result, we do not have any symmetry of type $^c \chi$ in (24). On the other hand, the state is unobservable. In particular, the orthogonal distribution $\mathcal{O}^\perp$ has a single generator. From (26), (A.4), and (A.7), we trivially obtain:

$$^u \chi_1 = \frac{T_I \delta(\delta - \rho)}{T_U V} - \frac{\delta(T_I + T_U)}{T_U}. \quad (\text{A.8})$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{l} \frac{dT_U'}{d\tau} = T_I' \delta' \\ \frac{dT_I'}{d\tau} = -T_I' \delta' \\ \frac{dV'}{d\tau} = 0 \\ \frac{d\lambda'}{d\tau} = 0 \\ \frac{d\rho'}{d\tau} = 0 \\ \frac{d\delta'}{d\tau} = \delta'(\delta' - \rho') \\ \frac{dN'}{d\tau} = N' \rho' \\ \frac{dc'}{d\tau} = 0 \\ \frac{d\eta'}{d\tau} = \frac{T_I' \delta'(\delta' - \rho')}{T_U' V'} - \frac{\delta'(T_I' + T_U')}{T_U'} \eta' \\ T_U'(t, 0) = T_U(t), \quad T_I'(t, 0) = T_I(t), \\ V'(t, 0) = V(t), \\ \lambda'(0) = \lambda, \quad \rho'(0) = \rho, \quad \delta'(0) = \delta, \quad N'(0) = N, \\ c'(0) = c, \quad \eta'(t, 0) = \eta(t) \end{array} \right. \quad (\text{A.9})$$

Its solution is (33), as it can be verified by substitution.

A.4. Analytical proof

We provide a direct proof of the indistinguishability of the states and parameters given in (33).

First of all, they provide the same outputs (trivially, $V'(t, \tau) = V(t)$ and $T_U'(t, \tau) + T_I'(t, \tau) = T_U(t) + T_I(t)$, $\forall \tau$).

In addition, they provide the same dynamics. In particular, from (1) and (33) it is possible to check, by a simple substitution, the validity of the following dynamics of the new three states (see all the details in Appendix A of Martinelli (2023a)):

$$\left\{ \begin{array}{l} \frac{d}{dt} T_U'(t, \tau) = \lambda' - \rho' T_U'(t, \tau) - \eta'(t, \tau) T_U'(t, \tau) V'(t) \\ \frac{d}{dt} T_I'(t, \tau) = \eta'(t, \tau) T_U'(t, \tau) V'(t) - \delta' T_I'(t, \tau) \\ \frac{d}{dt} V'(t) = N' \delta' T_I'(t, \tau) - c' V'(t) \end{array} \right. \quad (\text{A.10})$$

In other words, the new three states, satisfy exactly the same dynamical equations in (1) with the new parameters. We conclude that $T_U', T_I', V', \eta', \lambda', \delta', \rho', N'$, and c' cannot be distinguished from $T_U, T_I, V, \eta, \lambda, \delta, \rho, N$, and c , respectively. Regarding the model parameters, as $\eta' \neq \eta$, $\delta' \neq \delta$, and $N' \neq N$, we conclude that η , δ , and N are not identifiable. Finally, we remark that τ can take infinitesimal values and that, in the limit of $\tau \rightarrow 0$, we have: $T_U' \rightarrow T_U$, $T_I' \rightarrow T_I$, $V' \rightarrow V$, $\eta' \rightarrow \eta$, $\delta' \rightarrow \delta$, and $N' \rightarrow N$. Therefore, we conclude that the three parameters η , δ , and N are not even locally identifiable.

Appendix B. Analytical computations for Variant 2

B.1. Implementation of Algorithm 1

We run Algorithm 1 to obtain the observability codistribution. To proceed, we need, first of all, to introduce a state that includes both T_U, T_I, V , and all the constant parameters (i.e., $\lambda, \rho, \eta, \delta, c$). We set:

$$x = [T_U, T_I, V, \lambda, \rho, \eta, \delta, c]^T \quad (\text{B.1})$$

The system is directly a special case of (3). In particular, $m_u = 0$, $m_w = 1$, $p = 2$, $h_1(x) = V$, $h_2(x) = T_U + T_I$,

$$g^0 = \begin{bmatrix} \lambda - \rho T_U - \eta T_U V \\ \eta T_U V - \delta T_I \\ -c V \\ 0_5 \end{bmatrix}, \quad g^1 = \begin{bmatrix} 0 \\ 0 \\ \delta T_I \\ 0_5 \end{bmatrix} \quad (\text{B.2})$$

Line 2 provides:

$$\Omega = \text{span}\{\nabla h_1, \nabla h_2\} =$$

$$\text{span}\{[0, 0, 1, 0, 0, 0, 0, 0], [1, 1, 0, 0, 0, 0, 0, 0]\}.$$

We have: $\deg_w(\Omega) = 1 (= m_w)$, as $\mathcal{L}_{g^1} h_1 = \delta T_I \neq 0$. The system is in canonical form with respect to its unknown input.

The execution of Algorithm 1 continues with Line 45. The operation at this line ($[\tilde{h}_1, \dots, \tilde{h}_{m_w}] = S(\Sigma, \Omega)$) sets

$$\tilde{h}_1 = h_1 = V. \quad (\text{B.3})$$

Then, Line 46 provides:

$$\mu_0^0 = 1, \quad \mu_0^1 = 0, \quad \mu_1^0 = -c V, \quad \mu_1^1 = \delta T_I, \quad v_0^0 = 1, \quad v_0^1 = 0,$$

$$v_1^0 = \frac{c V}{T_I \delta}, \quad v_1^1 = \frac{1}{T_I \delta}, \quad (\text{B.4})$$

obtained from (4). In addition, from (5) we obtain:

$$\hat{g}^0 = \begin{bmatrix} \lambda - \rho T_U - \eta T_U V \\ \eta T_U V - \delta T_I \\ 0 \\ 0_5 \end{bmatrix}, \quad \hat{g}^1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0_5 \end{bmatrix}$$

Finally, from Equation (20) in [Martinelli \(2022\)](#) we obtain that, when $m_u = 0$, the codistribution $\tilde{\mathcal{O}}$ vanishes. The final step of Algorithm 1 is the execution of Line 47, namely:

$$\mathcal{O} = \left\langle \hat{g}^0, \hat{g}^1 \mid \text{span} \{ \nabla h_1, \nabla h_2 \} \right\rangle.$$

The algorithm in (1.45) in [Isidori \(1995\)](#) converges at the fifth step ($\Omega_5 = \Omega_4$), and

$$\mathcal{O} = \text{span} \left\{ \nabla h_1, \nabla h_2, \nabla \mathcal{L}_{\hat{g}^0} h_2, \nabla \mathcal{L}_{\hat{g}^0}^2 h_2, \nabla \mathcal{L}_{\hat{g}^0}^3 h_2, \nabla \mathcal{L}_{\hat{g}^1} \mathcal{L}_{\hat{g}^0}^2 h_2, \nabla \mathcal{L}_{\hat{g}^1}^4 h_2 \right\}. \quad (\text{B.5})$$

Its dimension is 7, which is smaller than the dimension of the state in (B.1). Hence, this state is not observable. We wish to determine the observability properties of T_U , T_I , and V , i.e., the first three components of the state in (B.1). We compute the distribution orthogonal to the observability codistribution in (B.5). We have:

$$\mathcal{O}^\perp = \text{span} \left\{ \begin{bmatrix} 0_7 \\ 1 \end{bmatrix} \right\}. \quad (\text{B.6})$$

B.2. Implementation of Algorithm 4

The first loop (Lines 2–8) checks the identifiability of the five constant parameters. It is immediate to check that one of them (c) is unidentifiable (it suffices to show that its gradient, i.e., the covector $[0, 0, 0, 0, 0, 0, 1]$ is not orthogonal to $\mathcal{O}^\perp$ and, consequently, it does not belong to $\mathcal{O}$). The remaining constant parameters, λ , ρ , η and δ , are locally identifiable.

The condition at Line 9 is not satisfied (the system is canonic and $m = m_w = 1$). The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (B.4). Line 14 provides the distribution given in (B.6).

The loop at Lines 15–18 consists of the single value $i_s = 1$. For this value, i.e., for the generator of $(\mathcal{O})^\perp$ in (B.6), we compute the coefficients ξ_i^α , for $\alpha = 0, 1$, and $i = 1$, by using (19), (B.2), (B.3), and (B.6). We have:

$$\mathcal{L}_{\hat{g}^0} \tilde{h}_1 = -Vc, \quad \mathcal{L}_{\hat{g}^1} \tilde{h}_1 = T_I \delta.$$

and

$$\Xi_1^{0,1} = \xi_1^0 = -V, \quad \Xi_1^{1,1} = \xi_1^1 = 0. \quad (\text{B.7})$$

The quantity in (20) for $\alpha = 0$ becomes:

$$C_1^{0,1} = \sum_{i=1}^m v_i^1 \Xi_i^{0,1} = v_1^1 \Xi_1^{0,1} = -\frac{V}{T_I \delta} \neq 0$$

where we used the expression of v_1^1 in (B.4).

As a result, the condition at Line 20 is not honoured, meaning that the time varying parameter $N(t)$ is not locally identifiable.

B.3. Application of the method in Sections 2.6 and 2.7

The orthogonal distribution $\mathcal{O}^\perp$ has a single generator. From (26), (B.4), and (B.7), we trivially obtain:

$${}^u \chi_1 = \frac{V}{T_I \delta}. \quad (\text{B.8})$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{ll} \frac{dT_U'}{d\tau} &= 0 \\ \frac{dT_I'}{d\tau} &= 0 \\ \frac{dV'}{d\tau} &= 0 \\ \frac{d\lambda'}{d\tau} &= 0 \\ \frac{d\rho'}{d\tau} &= 0 \\ \frac{d\eta'}{d\tau} &= 0 \\ \frac{d\delta'}{d\tau} &= 0 \\ \frac{dc'}{d\tau} &= 1 \\ \frac{dN'}{d\tau} &= \frac{V'}{T_I' \delta'} \\ T_U'(t, 0) &= T_U(t), \quad T_I'(t, 0) = T_I(t), \\ V'(t, 0) &= V(t), \\ \lambda'(0) = \lambda, \quad \rho'(0) = \rho, \quad \eta'(0) = \eta, \quad \delta'(0) = \delta, \quad c'(0) = c \\ N'(t, 0) &= N(t) \end{array} \right. \quad (\text{B.9})$$

Its solution is (37), as it can be verified by substitution.

We have obtained the following fundamental (new) result. For any choice of τ , the parameters c' and N' given in (37) are indistinguishable from c and N .

The validity of the above result can be directly obtained by proceeding as in [Appendix A.4](#).

Appendix C. Analytical computations for Variant 3

C.1. Implementation of Algorithm 1

We run Algorithm 1 to obtain the observability codistribution. To proceed, we need, first of all, to introduce a state that includes both T_U , T_I , V , and all the constant parameters (i.e., λ , ρ , δ , c). We set:

$$x = [T_U, T_I, V, \lambda, \rho, \delta, c]^T \quad (\text{C.1})$$

The system is directly a special case of (3). In particular, $m_u = 0$, $m_w = 2$, $p = 3$, $h_1(x) = V$, $h_2(x) = T_U$, $h_3(x) = T_I$,

$$g^0 = \begin{bmatrix} \lambda - \rho T_U \\ -\delta T_I \\ -cV \\ 0_4 \end{bmatrix}, \quad g^1 = \begin{bmatrix} -T_U V \\ T_U V \\ 0 \\ 0_4 \end{bmatrix}, \quad g^2 = \begin{bmatrix} 0 \\ 0 \\ \delta T_I \\ 0_4 \end{bmatrix} \quad (\text{C.2})$$

Line 2 provides:

$$\Omega = \text{span} \{ \nabla h_1, \nabla h_2, \nabla h_3 \} =$$

$$\text{span} \{ [0, 0, 1, 0, 0, 0, 0], [1, 0, 0, 0, 0, 0, 0], [0, 1, 0, 0, 0, 0, 0] \}.$$

Let us compute $\deg_w(\Omega)$. The reconstructability matrix from h_1 , h_2 is:

$$\mathcal{RM}(h_1, h_2) = \begin{bmatrix} \mathcal{L}_{g^1} h_1 & \mathcal{L}_{g^2} h_1 \\ \mathcal{L}_{g^1} h_2 & \mathcal{L}_{g^2} h_2 \end{bmatrix} = \begin{bmatrix} 0 & T_I \delta \\ -T_U V & 0 \end{bmatrix}$$

Its rank is 2. Hence: $\deg_w(\Omega) = 2 (= m_w)$, and the system is in canonical form with respect to its unknown input.

The execution of Algorithm 1 continues with Line 45. The operation at this line ($[\tilde{h}_1, \dots, \tilde{h}_{m_w}] = S(\Sigma, \Omega)$) sets

$$\tilde{h}_1 = h_1 = V, \quad \tilde{h}_2 = h_2 = T_U. \quad (\text{C.3})$$

Then, Line 46 provides:

$$\mu_0^0 = 1, \quad \mu_0^1 = \mu_0^2 = \mu_2^2 = \mu_1^1 = 0, \quad \mu_1^0 = -cV,$$

$$\mu_1^2 = T_I \delta, \mu_2^0 = \lambda - T_U \rho, \mu_2^1 = -T_U V,$$

$$v_0^0 = 1, v_0^1 = v_0^2 = v_1^2 = v_1^1 = 0,$$

$$v_1^0 = \frac{\lambda - T_U \rho}{T_U V}, v_1^1 = -\frac{1}{T_U V}, v_2^0 = \frac{cV}{T_I \delta}, v_2^1 = \frac{1}{T_I \delta}, \quad (C.4)$$

obtained from (4). In addition, from (5) we obtain:

$$\hat{g}^0 = \begin{bmatrix} 0 \\ \lambda - T_I \delta - T_U \rho \\ 0 \\ 0_4 \end{bmatrix}, \quad \hat{g}^1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0_4 \end{bmatrix}, \quad \hat{g}^2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0_4 \end{bmatrix}$$

Finally, from Equation (20) in Martinelli (2022) we obtain that, when $m_u = 0$, the codistribution $\tilde{\mathcal{O}}$ vanishes. The final step of Algorithm 1 is the execution of Line 47, namely:

$$\mathcal{O} = \langle \hat{g}^0, \hat{g}^1, \hat{g}^2 | \text{span} \{ \nabla h_1, \nabla h_2, \nabla h_3 \} \rangle.$$

The algorithm in (1.45) in Isidori (1995) converges at the third step ($\Omega_3 = \Omega_2$), and $\mathcal{O} =$

$$\text{span} \{ \nabla h_1, \nabla h_2, \nabla h_3, \nabla \mathcal{L}_{\hat{g}^0} h_3, \nabla \mathcal{L}_{\hat{g}^0}^2 h_3, \nabla \mathcal{L}_{\hat{g}^2} \mathcal{L}_{\hat{g}^0} h_3 \}. \quad (C.5)$$

The orthogonal distribution is:

$$\mathcal{O}^\perp = \text{span} \left\{ \begin{bmatrix} 0_6 \\ 1 \end{bmatrix} \right\}, \quad (C.6)$$

C.2. Implementation of Algorithm 4

The first loop (Lines 2–8) checks the identifiability of the four constant parameters. It is immediate to check that one of them (c) is unidentifiable, exactly as in the second variant studied in Section 4. The remaining constant parameters, λ , ρ , and δ , are locally identifiable.

The condition at Line 9 is not satisfied (the system is canonic and $m = m_w = 2$). The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (C.4). Line 14 provides the distribution given in (C.6).

The loop at Lines 15–18 consists of the single value $i_s = 1$. For this value, i.e., for the generator of $(\mathcal{O})^\perp$ in (C.6), we compute the coefficients ξ_i^α , for $\alpha = 0, 1, 2$, and $i = 1, 2$, by using (19), (C.2), (C.3), and (C.6). We have:

$$\mathcal{L}_{g^0} \tilde{h}_1 = -Vc, \quad \mathcal{L}_{g^1} \tilde{h}_1 = 0, \quad \mathcal{L}_{g^2} \tilde{h}_1 = \delta T_I, \quad \mathcal{L}_{g^0} \tilde{h}_2 = \lambda - \rho T_U,$$

$$\mathcal{L}_{g^1} \tilde{h}_2 = -T_U V, \quad \mathcal{L}_{g^2} \tilde{h}_2 = 0.$$

and

$$\Xi_1^{0,1} = \xi_1^0 = -V, \quad \Xi_2^{0,1} = \xi_2^0 = 0, \quad \Xi_1^{1,1} = \xi_1^1 = 0, \quad (C.7)$$

$$\Xi_2^{1,1} = \xi_2^1 = 0, \quad \Xi_1^{2,1} = \xi_1^2 = 0, \quad \Xi_2^{2,1} = \xi_2^2 = 0.$$

Let us consider the first time varying parameter. We must set $j = 1$ and compute the quantities in (20) for $\alpha = 0, 1, 2$. By using (C.4) we obtain, $\forall \alpha$, $C_1^{\alpha,1} =$

$$\sum_{i=1}^m v_1^i \Xi_i^{\alpha,1} = \sum_{i=1}^2 v_1^i \Xi_i^{\alpha,1} = v_1^2 \Xi_2^{\alpha,1} = -\frac{1}{T_U V} \Xi_2^{\alpha,1} = 0,$$

as $\Xi_2^{\alpha,1} = 0 \forall \alpha$. As a result, the condition at Line 20 is certainly honoured for $j = 1$, meaning that the first time varying parameter $\eta(t)$ is locally identifiable.

Let us consider the second time varying parameter. We must set $j = 2$ and compute the quantities in (20) for $\alpha = 0, 1, 2$. By using (C.4) we obtain:

$$C_2^{\alpha,1} = \sum_{i=1}^m v_2^i \Xi_i^{\alpha,1} = \sum_{i=1}^2 v_2^i \Xi_i^{\alpha,1} = v_2^1 \Xi_1^{\alpha,1} = \frac{1}{T_I \delta} \Xi_1^{\alpha,1}$$

which is $-\frac{V}{T_I \delta} \neq 0$, when $\alpha = 0$. As a result, the condition at Line 20 is not honoured for $j = 2$, meaning that the second time varying parameter $N(t)$ is not locally identifiable.

C.3. Application of the method in Sections 2.6 and 2.7

The orthogonal distribution $\mathcal{O}^\perp$ has a single generator. From (26), (C.4), and (C.7), we trivially obtain:

$${}^u \chi_1 = 0, \quad {}^u \chi_2 = \frac{V}{T_I \delta}. \quad (C.8)$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{ll} \frac{dT_U'}{d\tau} &= 0 \\ \frac{dT_I'}{d\tau} &= 0 \\ \frac{dV'}{d\tau} &= 0 \\ \frac{d\lambda'}{d\tau} &= 0 \\ \frac{d\rho'}{d\tau} &= 0 \\ \frac{d\delta'}{d\tau} &= 0 \\ \frac{dc'}{d\tau} &= 1 \\ \frac{d\eta'}{d\tau} &= 0 \\ \frac{dN'}{d\tau} &= \frac{V'}{T_I \delta'} \\ T_U'(t, 0) &= T_U(t), \quad T_I'(t, 0) = T_I(t), \\ V'(t, 0) &= V(t), \\ \lambda'(0) = \lambda, \quad \rho'(0) &= \rho, \quad \delta'(0) = \delta, \quad c'(0) = c \\ \eta'(t, 0) &= \eta(t) \\ N'(t, 0) &= N(t) \end{array} \right. \quad (C.9)$$

Its solution is (40), as it can be verified by substitution.

We have obtained the following fundamental (new) result. For any choice of τ , the parameters c' and N' given in (40) are indistinguishable from c and N .

The validity of the above result can be directly obtained by proceeding as in Appendix A.4.

Appendix D. Analytical computations for Variant 4

D.1. Implementation of Algorithm 1

We run Algorithm 1 to obtain the observability codistribution. To proceed, we need, first of all, to introduce a state that includes both T_U , T_I , V , and all the constant parameters (i.e., λ , ρ , δ , c). We obtain exactly the same state given in (C.1). We also obtain exactly the same vector fields that characterize the dynamics for the third variant studied in Section 5. The only difference is that we have now 2 outputs instead of 3. Specifically, we have: $h_1(x) = V$, $h_2(x) = T_U + T_I$.

Line 2 provides:

$$\Omega = \text{span} \{ \nabla h_1, \nabla h_2 \} =$$

$$\text{span} \{ [0, 0, 1, 0, 0, 0, 0], [1, 1, 0, 0, 0, 0, 0] \}.$$

Let us compute $\deg_w(\Omega)$. The reconstructability matrix from h_1, h_2 is:

$$\mathcal{RM}(h_1, h_2) = \begin{bmatrix} \mathcal{L}_{g^1} h_1 & \mathcal{L}_{g^2} h_1 \\ \mathcal{L}_{g^1} h_2 & \mathcal{L}_{g^2} h_2 \end{bmatrix} = \begin{bmatrix} 0 & T_I \delta \\ 0 & 0 \end{bmatrix}$$

Its rank is 1. Hence: $\deg_w(\Omega) = 1 (< m_w)$, and the system is not in canonical form with respect to its unknown input. We enter the loop at lines 9–14.

Line 10 provides $m = 1$ and Line 11 provides $\tilde{h}_1 = h_1$. In addition, as $\mathcal{L}_{g^1} \tilde{h}_1 = 0$ and $\mathcal{L}_{g^2} \tilde{h}_1 \neq 0$ the unknown inputs are reordered, i.e., g^2 becomes the old g^1 and g^1 becomes the old g^2 . In other words, Σ will be characterized by the vector fields:

$$g^0 = \begin{bmatrix} \lambda - \rho T_U \\ -\delta T_I \\ -cV \\ 0_4 \end{bmatrix}, \quad g^1 = \begin{bmatrix} 0 \\ 0 \\ \delta T_I \\ 0_4 \end{bmatrix}, \quad g^2 = \begin{bmatrix} -T_U V \\ T_U V \\ 0 \\ 0_4 \end{bmatrix} \quad (D.1)$$

(which provide $\mathcal{L}_{g^1} \tilde{h}_1 = T_I \delta \neq 0$ and $\mathcal{L}_{g^2} \tilde{h}_1 = 0$), and $w_1 = N$, and $w_2 = \eta$. Line 12 provides:

$$\mu_0^0 = 1, \mu_0^1 = 0, \mu_1^0 = -cV, \mu_1^1 = \delta T_I,$$

$$\nu_0^0 = 1, \nu_0^1 = 0, \nu_1^0 = \frac{cV}{T_I \delta}, \nu_1^1 = \frac{1}{T_I \delta},$$

obtained from (4). In addition, from (5) we obtain:

$$\hat{g}^0 = \begin{bmatrix} \lambda - \rho T_U \\ -\delta T_I \\ 0 \\ 0_4 \end{bmatrix}, \quad \hat{g}^1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0_4 \end{bmatrix}.$$

Line 13 computes

$$\Omega = \left\langle \left\{ \begin{bmatrix} \hat{g}^0 \\ \hat{g}^1 \end{bmatrix} \right\} \mid \text{span} \{ \nabla h_1, \nabla h_2 \} \right\rangle.$$

By a direct computation, we obtain:

$$\Omega = \text{span} \{ \nabla h_1, \nabla h_2, \nabla \mathcal{L}_{g^0} h_2 \}.$$

Now, $\deg_w(\Omega) = 2 = m_w$. Hence, the second condition at Line 9 is not met and the loop is interrupted. In addition, as the condition at Line 15 is met, also the main loop is interrupted. The execution of Algorithm 1 continues with Line 45. The operation at this line ($[\tilde{h}_1, \dots, \tilde{h}_{m_w}] = S(\Sigma, \Omega)$) sets

$$\tilde{h}_1 = h_1 = V, \quad \tilde{h}_2 = \mathcal{L}_{g^0} h_2 = \lambda - \rho T_U - \delta T_I. \quad (D.2)$$

Then, Line 46 provides:

$$\mu_0^0 = 1, \mu_0^1 = \mu_0^2 = \mu_1^2 = \mu_2^2 = 0, \mu_1^0 = -cV, \mu_1^1 = T_I \delta,$$

$$\mu_2^0 = T_I \delta^2 - \rho(\lambda - T_U \rho), \mu_2^1 = T_U V(\rho - \delta),$$

$$\nu_0^0 = 1, \nu_0^1 = \nu_0^2 = \nu_1^2 = \nu_2^2 = 0, \nu_1^0 = \frac{cV}{T_I \delta}, \nu_1^1 = \frac{1}{T_I \delta}, \quad (D.3)$$

$$\nu_2^0 = \frac{T_I \delta^2 + T_U \rho^2 - \lambda \rho}{T_U V(\delta - \rho)}, \nu_2^1 = \frac{1}{T_U V(\rho - \delta)},$$

obtained from (4). In addition, from (5), (D.1), and (D.3) we obtain:

$$\hat{g}^0 = \begin{bmatrix} \frac{-\delta(T_I \delta - \lambda + T_U \rho)}{\delta - \rho} \\ \frac{\rho(T_I \delta - \lambda + T_U \rho)}{\delta - \rho} \\ 0 \\ 0_4 \end{bmatrix}, \quad \hat{g}^1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0_4 \end{bmatrix}, \quad \hat{g}^2 = \begin{bmatrix} 1/(\delta - \rho) \\ -1/(\delta - \rho) \\ 0 \\ 0_4 \end{bmatrix},$$

Finally, from Equation (20) in Martinelli (2022) we obtain that, when $m_u = 0$, the codistribution $\tilde{\mathcal{O}}$ vanishes. The final step of Algorithm 1 is the execution of Line 47, namely:

$$\mathcal{O} = \left\langle \hat{g}^0, \hat{g}^1, \hat{g}^2 \mid \text{span} \{ \nabla h_1, \nabla h_2, \nabla \tilde{h}_2 \} \right\rangle.$$

The algorithm in (1.45) in Isidori (1995) converges at the first step ($\Omega_1 = \Omega_0$), and

$$\mathcal{O} = \text{span} \{ \nabla h_1, \nabla h_2, \nabla \tilde{h}_2 \}. \quad (D.4)$$

Its dimension is 3, which is smaller than the dimension of the state in (C.1). Hence, this state is not observable. We wish to determine the observability properties of T_U, T_I , and V , i.e., the first three components of the state in (C.1). We compute the distribution orthogonal to the observability codistribution in (D.4). We have:

$$\mathcal{O}^\perp = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ \delta - \rho \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} T_U \\ -T_U \\ 0 \\ \delta - \rho \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} T_I \\ -T_I \\ 0 \\ 0 \\ \delta - \rho \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad (D.5)$$

D.2. Implementation of Algorithm 4

The first loop (Lines 2–8) checks the identifiability of the four constant parameters.

It is immediate to check that no one is locally identifiable. Indeed, their gradients are not orthogonal to $\mathcal{O}^\perp$ (e.g., for λ , the covector $\nabla \lambda = [0, 0, 0, 1, 0, 0, 0]$ is not orthogonal to the first generator of $\mathcal{O}^\perp$). As a result, all these gradients cannot belong to $\mathcal{O}$ (e.g., $\nabla \lambda \notin \mathcal{O}$).

The condition at Line 9 is not satisfied (the system is canonic and $m = m_w = 2$). The execution of Algorithm 4 continues with Lines 12 and 13, which compute the tensors given in (D.3). Line 14 provides the distribution given in (D.5).

The loop at Lines 15–18 consists of the four values $i_s = 1, 2, 3, 4$.

Let us consider the first time varying parameter, i.e., $N(t)$.¹⁷ We must set $j = 1$ and check the condition at Line 20 for $\alpha = 0, 1, 2$ and for all the generators of $\mathcal{O}^\perp$ ($i_s = 1, 2, 3, 4$).

We have:

$$C_1^{\alpha, i_s} = \sum_{i=1}^2 v_i^i \Xi_i^{\alpha, i_s} = v_1^1 \Xi_1^{\alpha, i_s} = \frac{1}{T_I \delta} \Xi_1^{\alpha, i_s},$$

where we used the expression of $v_1^1 = \frac{1}{T_I \delta}$ and $v_1^2 = 0$ given in (D.3). The expression of Ξ_1^{α, i_s} is given (19). It depends on the generator of $\mathcal{O}^\perp$ and on α . In particular, it suffices to find one generator of $\mathcal{O}^\perp$, and one value of α among 0, 1, 2, such that $\Xi_1^{\alpha, i_s} \neq 0$, to conclude that the condition at Line 20 is not honoured and, as a result, that the parameter $N(t)$ is not locally identifiable. Let us set $\alpha = 0$ and let us consider the fourth generator in (D.5) ($i_s = 4$). From (D.1) and (D.2), we have $\mathcal{L}_{g^0} \tilde{h}_1 = -cV$ and, consequently:

$$\Xi_1^{0,4} = \left(\frac{\partial}{\partial x} \mathcal{L}_{g^0} \tilde{h}_1 \right) \cdot [0, 0, 0, 0, 0, 0, 1]^T =$$

$$\left(\frac{\partial}{\partial x} (-cV) \right) \cdot [0, 0, 0, 0, 0, 0, 1]^T = -V \neq 0.$$

Therefore, the parameter $N(t)$ is not locally identifiable.

¹⁷ After the execution of Algorithm 1, the two UIs have been reordered. In other words, the system $\mathcal{E}$ built by Algorithm 1 is characterized by the two unknown inputs $w_1 = N$ and $w_2 = \eta$.

Let us consider the second time varying parameter, i.e., $\eta(t)$. We must set $j = 2$ and check the condition at Line 20 for $\alpha = 0, 1, 2$ and for all the generators of $\mathcal{O}^\perp$ ($i_s = 1, 2, 3, 4$). We have:

$$C_2^{\alpha, i_s} = \sum_{i=1}^2 v_2^i \Xi_i^{\alpha, i_s} = v_2^2 \Xi_2^{\alpha, i_s} = \frac{1}{T_U V(\rho - \delta)} \Xi_2^{\alpha, i_s},$$

where we used the expression of $v_2^2 = \frac{1}{T_U V(\rho - \delta)}$ and $v_2^1 = 0$ given in (D.3). The expression of Ξ_2^{α, i_s} is given (19). It depends on the generator of $\mathcal{O}^\perp$ and on α . In particular, it suffices to find one generator of $\mathcal{O}^\perp$, and one value of α among 0, 1, 2, such that $\Xi_2^{\alpha, i_s} \neq 0$, to conclude that the condition at Line 20 is not honoured and, as a result, that the parameter $\eta(t)$ is not locally identifiable. Let us set $\alpha = 2$ and let us consider the first generator in (D.5) ($i_s = 1$). From (D.1) and (D.2), we have $\mathcal{L}_{g^2} \tilde{h}_2 = T_U V(\rho - \delta)$ and, consequently:

$$\Xi_2^{2,1} = \left(\frac{\partial}{\partial x} \mathcal{L}_{g^2} \tilde{h}_2 \right) \cdot [-1, 1, 0, \delta - \rho, 0, 0, 0]^T = V(\delta - \rho) \neq 0.$$

Therefore, also the parameter $\eta(t)$ is not locally identifiable.

D.3. Application of the method in Sections 2.6 and 2.7

The orthogonal distribution $\mathcal{O}^\perp$ has four independent generators. Hence, we obtain four independent sets of values. In particular, each set of values is obtained by solving the differential equations system in (30) for every generator in (D.5).

D.3.1. First set

We consider the first generator in (D.5). From (19), (D.1), and (D.2), and by setting $\xi = [-1, 1, 0, \delta - \rho, 0, 0, 0]^T$, we obtain:

$$\xi_1^0 = \xi_2^1 = \xi_1^2 = 0, \quad \xi_2^0 = \delta(\delta - \rho), \quad \xi_1^1 = \delta, \quad \xi_2^2 = V(\delta - \rho). \quad (D.6)$$

From (26), (D.3), and (D.6), we trivially obtain:

$${}^u \chi_1 = -\frac{N}{T_I}, \quad {}^u \chi_2 = \frac{\delta + V\eta}{T_U V}. \quad (D.7)$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{ll} \frac{dT'_U}{d\tau} &= -1 \\ \frac{dT'_I}{d\tau} &= 1 \\ \frac{dV'}{d\tau} &= 0 \\ \frac{d\lambda'}{d\tau} &= \delta' - \rho' \\ \frac{d\rho'}{d\tau} &= 0 \\ \frac{d\delta'}{d\tau} &= 0 \\ \frac{dc'}{d\tau} &= 0 \\ \frac{dN'}{d\tau} &= -\frac{N'}{T'_I} \\ \frac{d\eta'}{d\tau} &= \frac{\delta' + V'\eta'}{T'_I V'} \\ T'_U(t, 0) &= T_U(t), \quad T'_I(t, 0) = T_I(t), \\ V'(t, 0) &= V(t), \\ \lambda'(0) = \lambda, \quad \rho'(0) &= \rho, \quad \delta'(0) = \delta, \quad c'(0) = c \\ N'(t, 0) &= N(t) \\ \eta'(t, 0) &= \eta(t) \end{array} \right. \quad (D.8)$$

Its solution is (41), as it can be verified by substitution.

We have obtained the following new result. For any choice of τ , the state and the parameters given in (41) are indistinguishable from the true state and the true parameters. This could also be directly verified

by showing that they provide the same outputs (trivially, $V'(t, \tau) = V(t)$ and $T'_U(t, \tau) + T'_I(t, \tau) = T_U(t) + T_I(t)$, $\forall \tau$) and that they provide the same dynamics.

D.3.2. Second set

We consider the second generator in (D.5). From (19), (D.1), and (D.2), and by setting $\xi = [T_U, -T_U, 0, 0, \delta - \rho, 0, 0]^T$, we obtain:

$$\xi_1^0 = \xi_2^1 = \xi_1^2 = \xi_2^2 = 0,$$

$$\xi_2^0 = (\rho - \delta)\lambda - (\rho - \delta)^2 T_U, \quad \xi_1^1 = -T_U \delta. \quad (D.9)$$

From (26), (D.3), and (D.9), we trivially obtain:

$${}^u \chi_1 = \frac{T_U N}{T_I}, \quad {}^u \chi_2 = -\frac{\lambda + T_U(\delta - \rho)}{T_U V}. \quad (D.10)$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{ll} \frac{dT'_U}{d\tau} &= T'_U \\ \frac{dT'_I}{d\tau} &= -T'_U \\ \frac{dV'}{d\tau} &= 0 \\ \frac{d\lambda'}{d\tau} &= 0 \\ \frac{d\rho'}{d\tau} &= \delta' - \rho' \\ \frac{d\delta'}{d\tau} &= 0 \\ \frac{dc'}{d\tau} &= 0 \\ \frac{dN'}{d\tau} &= \frac{T'_U N'}{T'_I} \\ \frac{d\eta'}{d\tau} &= -\frac{\lambda' + T'_U(\delta' - \rho')}{T'_I V'} \\ T'_U(t, 0) &= T_U(t), \quad T'_I(t, 0) = T_I(t), \\ V'(t, 0) &= V(t), \\ \lambda'(0) = \lambda, \quad \rho'(0) &= \rho, \quad \delta'(0) = \delta, \quad c'(0) = c \\ N'(t, 0) &= N(t) \\ \eta'(t, 0) &= \eta(t) \end{array} \right. \quad (D.11)$$

Its solution is (42), as it can be verified by substitution.

We have obtained the following new result. For any choice of τ , the state and the parameters given in (42) are indistinguishable from the true state and the true parameters. This could also be directly verified by showing that they provide the same outputs (trivially, $V'(t, \tau) = V(t)$ and $T'_U(t, \tau) + T'_I(t, \tau) = T_U(t) + T_I(t)$, $\forall \tau$) and that they provide the same dynamics.

D.3.3. Third set

We consider the third generator in (D.5). From (19), (D.1), and (D.2), and by setting $\xi = [T_I, -T_I, 0, 0, 0, \delta - \rho, 0]^T$, we obtain:

$$\xi_1^0 = \xi_2^1 = \xi_1^2 = 0, \quad \xi_2^0 = T_I(\delta - \rho)^2, \quad (D.12)$$

$$\xi_1^1 = -T_I \rho, \quad \xi_2^2 = -V(T_I + T_U)(\delta - \rho).$$

From (26), (D.3), and (D.12), we trivially obtain:

$${}^u \chi_1 = \frac{\rho N}{\delta}, \quad {}^u \chi_2 = \frac{T_I(\delta - \rho)^2 - V\eta(T_I + T_U)(\delta - \rho)}{T_U V(\delta - \rho)}. \quad (D.13)$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{l} \frac{dT'_U}{d\tau} = T'_I \\ \frac{dT'_I}{d\tau} = -T'_I \\ \frac{dV'}{d\tau} = 0 \\ \frac{d\lambda'}{d\tau} = 0 \\ \frac{d\rho'}{d\tau} = 0 \\ \frac{d\delta'}{d\tau} = \delta' - \rho' \\ \frac{dc'}{d\tau} = 0 \\ \frac{dN'}{d\tau} = N' \frac{\rho'}{\delta'} \\ \frac{d\eta'}{d\tau} = \frac{T'_I(\delta' - \rho')}{T'_U V'} - \eta' \frac{(T'_I + T'_U)}{T'_U} \\ T'_U(t, 0) = T_U(t), T'_I(t, 0) = T_I(t), \\ V'(t, 0) = V(t), \\ \lambda'(0) = \lambda, \rho'(0) = \rho, \delta'(0) = \delta, c'(0) = c \\ N'(t, 0) = N(t) \\ \eta'(t, 0) = \eta(t) \end{array} \right. \quad (D.14)$$

Its solution is (43), as it can be verified by substitution.

We have obtained the following new result. For any choice of τ , the state and the parameters given in (43) are indistinguishable from the true state and the true parameters. This could also be directly verified by showing that they provide the same outputs (trivially, $V'(t, \tau) = V(t)$ and $T'_U(t, \tau) + T'_I(t, \tau) = T_U(t) + T_I(t)$, $\forall \tau$) and that they provide the same dynamics.

D.3.4. Fourth set

We consider the fourth generator in (D.5). From (19), (D.1), and (D.2), and by setting $\xi = [0, 0, 0, 0, 0, 0, 1]^T$, we obtain:

$$\xi_1^0 = -V, \quad \xi_2^0 = \xi_1^1 = \xi_2^1 = \xi_1^2 = \xi_2^2 = 0. \quad (D.15)$$

From (26), (D.3), and (D.15), we trivially obtain:

$${}^u\chi_1 = \frac{V}{T_I \delta}, \quad {}^u\chi_2 = 0. \quad (D.16)$$

By an explicit computation, it is possible to check that the condition in (29) is honoured. The differential equations system in (30) becomes:

$$\left\{ \begin{array}{l} \frac{dT'_U}{d\tau} = 0 \\ \frac{dT'_I}{d\tau} = 0 \\ \frac{dV'}{d\tau} = 0 \\ \frac{d\lambda'}{d\tau} = 0 \\ \frac{d\rho'}{d\tau} = 0 \\ \frac{d\delta'}{d\tau} = 0 \\ \frac{dc'}{d\tau} = 1 \\ \frac{dN'}{d\tau} = \frac{V'}{T'_I \delta'} \\ \frac{d\eta'}{d\tau} = 0 \\ T'_U(t, 0) = T_U(t), T'_I(t, 0) = T_I(t), \\ V'(t, 0) = V(t), \\ \lambda'(0) = \lambda, \rho'(0) = \rho, \delta'(0) = \delta, c'(0) = c \\ N'(t, 0) = N(t) \\ \eta'(t, 0) = \eta(t) \end{array} \right. \quad (D.17)$$

Its solution is (44), as it can be verified by substitution.

References

- Basile, G., Marro, G., 1969. On the observability of linear, time invariant systems with unknown inputs. *J. Optim. Theory Appl.* 3, 410–415.
- Belo, F.A.W., Salaris, P., Bicchi, A., 2010. 3 known landmarks are enough for solving planar bearing SLAM and fully reconstruct unknown inputs. In: *IROS 2010*. Taipei, Taiwan.
- Casti, J.L., 1982. Recent developments and future perspectives in nonlinear system theory. *SIAM Rev.* 24 (3).
- Chatzis, Manolis N., Chatzi, Eleni N., Smyth, Andrew W., 2015. On the observability and identifiability of nonlinear structural and mechanical systems. *Struct. Control Health Monit.* 22, 574–593.
- Chen, J., Wu, H., 2008. Estimation of time-varying parameters in deterministic dynamic models. *Statist. Sinica* 18, 987–1006.
- Chis, O.-T., Banga, J.R., Balsa-Canto, E., 2011. Structural identifiability of systems biology models: A critical comparison of methods. *PLoS ONE* 6 (11).
- DiStefano, III, J., 2015. *Dynamic Systems Biology Modeling and Simulation*. Academic Press.
- Elias August 1, Papachristodoulou, Antonis, 2009. A new computational tool for establishing model parameter identifiability. *J. Comput. Biol.* 16 (6), 875–885.
- Grewal, M., Glover, K., 1976. Identifiability of linear and nonlinear dynamical systems. *IEEE Trans. Automat. Control* 21 (6), 833–837. <http://dx.doi.org/10.1109/TAC.1976.1101375>.
- Guidorzi, R., Marro, G., 1971. On wonham stabilizability condition in the synthesis of observers for unknown-input systems. *IEEE Trans. Automat. Control* 16 (5), 499–500.
- Guo, Chao X., Roumeliotis, Stergios I., Roumeliotis IMU-RGBD camera 3D pose estimation and extrinsic calibration: Observability analysis and consistency improvement. In: *2013 IEEE International Conference on Robotics and Automation*. pp. 2935–2942.
- Hartmann, A., Vinga, Susana, Lemos, J.M., 2012. Online Bayesian time-varying parameter estimation of HIV-1 time-series. In: *16th IFAC Symposium on System Identification* Brussels, Belgium, pp. 11–13.
- Hermann, R., Krener, A.J., 1977. Nonlinear controllability and observability. *IEEE Trans. Automat. Control* AC-22 (5), 728–740.
- Hesch, Joel A., Kottas, Dimitris G., Bowman, Sean L., Roumeliotis, Stergios I., 2014. Camera-IMU-based localization: Observability analysis and consistency improvement. *Int. J. Robot. Res.* 33 (1), 182–201.
- Ho, D.D., Neumann, A.U., Perelson, A.S., Chen, W., Leonard, J.M., Markowitz, M., 1995. Rapid turnover of plasma virions and CD4 lymphocytes in HIV-1 infection. *Nature* 373, 123–126.
- Isidori, A., 1995. *Nonlinear Control Systems*, third ed. Springer Verlag, London.
- Kalman, R.E., 1961. On the general theory of control systems. In: *Proc. 1st Int. Cong. of IFAC*. Moscow 1960 1481, Butterworth, London.
- Kalman, R.E., 1963. Mathematical description of linear dynamical systems. *SIAM J. Control* 1, 152.
- Maes, K., Chatzis, M., Lombaert, G., 2019. Observability of nonlinear systems with unmeasured inputs. *Mech. Syst. Signal Process.* 130, 378–394.
- Martinelli, A., 2015. Extension of the observability rank condition to nonlinear systems driven by unknown inputs. In: *23th Mediterranean Conference on Control and Automation*. MED, Torremolinos, Spain.
- Martinelli, A., 2019. Nonlinear unknown input observability: Extension of the observability rank condition. *IEEE Trans. Automat. Control* 64 (1), 222–237. <http://dx.doi.org/10.1109/TAC.2018.2798806>.
- Martinelli, A., 2020. Observability: A new theory based on the group of invariance. In: *SIAM in Advances in Design and Control*. ISBN: 978-1-61197-624-3.
- Martinelli, A., 2022. Nonlinear unknown input observability and unknown input reconstruction: The general analytical solution. *Inf. Fusion* (ISSN: 1566-2535) 85, 23–51. <http://dx.doi.org/10.1016/j.inffus.2022.03.004>, <https://www.sciencedirect.com/science/article/pii/S156625352200029X>.
- Martinelli, A., 2023a. Detection of a very serious error in the paper: On identifiability of nonlinear ODE models and applications in viral dynamics, [arXiv: 2304.14288\[math.OA\]](https://arxiv.org/abs/2304.14288). A newer and more complete version is available at <https://inria.hal.science/hal-04419008>.
- Martinelli, A., 2023b. General analytical condition to nonlinear identifiability and its application in viral dynamics, [arXiv:2305.01060\[math.OA\]](https://arxiv.org/abs/2305.01060).
- Martinelli, A., 2023c. Identifiability of nonlinear ODE Models with time varying Parameters: the General Analytical Solution and Applications in Viral Dynamics, [arXiv:2211.13507\[ess.SY\]](https://arxiv.org/abs/2211.13507).
- Mhawej, M.J., Moog, C.H., Biafore, F., Brunet-François, C., 2010. Control of the HIV infection and drug dosage. *Biomed. Signal Process. Control* (ISSN: 1746-8094) 5 (1), 45–52. <http://dx.doi.org/10.1016/j.bspc.2009.05.001>.
- Miao, H., Perelson, A.S., Wu, H., 2023. Erratum: On identifiability of nonlinear ODE models and applications in viral dynamics. *SIAM Rev.* 65 (3), 732.
- Miao, H., Xia, X., Perelson, A.S., Wu, H., 2011. On identifiability of nonlinear ODE models and applications in viral dynamics. *SIAM Rev.* 53 (1), 3–39.
- Paul, M.K., Roumeliotis, S.I., 2018. Alternating-stereo VINS: Observability analysis and performance evaluation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 4729–4737.

- Perelson, A.S., 2002. Modelling viral and immune system dynamics. *Nat. Rev. Immunol.* 2, 28–36.
- Perelson, A.S., Essunger, P., Cao, Y., Vesanen, M., Hurley, A., Saksela, K., Markowitz, M., Ho, D.D., 1997. Decay characteristics of HIV-1-infected compartments during combination therapy. *Nature* 387, 188–191.
- Perelson, A.S., Kirschner, D., 1993. Dynamics of HIV infection of CD4+ T cells. *Math. Biosci.* 114, 81–125.
- Perelson, A.S., Nelson, P.W., 1999. Mathematical analysis of HIV-1 dynamics in vivo. *SIAM Rev.* 41, 3–44.
- Sarafrazi, M.A., Kotta, U., Bartosiewicz, Z., 2022. On the stopping criteria in nonlinear unknown input observability condition. *IEEE Trans. Automat. Control* 67 (9), 4733–4737. <http://dx.doi.org/10.1109/TAC.2022.3160133>.
- Schiano, Fabrizio, Tron, Roberto, 2018. The dynamic bearing observability matrix nonlinear observability and estimation for multi-agent systems. In: 2018 IEEE International Conference on Robotics and Automation.
- Tunali, E., Tarn, Tzyh-Jong, 1987. New results for identifiability of nonlinear systems. *IEEE Trans. Automat. Control* 32 (2), 146–154. <http://dx.doi.org/10.1109/TAC.1987.1104544>.
- Villaverde, Alejandro F., 2019. Observability and structural identifiability of nonlinear biological systems. *Complexity Special Issue on Computational Methods for Identification and Modelling of Complex Biological Systems*.
- Villaverde, A.F., Barreiro, A., 2016. Identifiability of large nonlinear biochemical networks. *MATCH Commun. Math. Comput. Chem.* 76 (2), 259–296.
- Villaverde, A., Barreiro, A.F., Papachristodoulou, A., 2016. Structural identifiability of dynamic systems biology models. *PLoS Comput. Biol.* <http://dx.doi.org/10.1371/journal.pcbi.1005153>.
- Villaverde, A.F., Tsiantis, N., Banga, J.R., 2019. Full observability and estimation of unknown inputs, states, and parameters of nonlinear biological models. *J. R. Soc. Interface* 16.
- Walter, E., Pronzato, L., 1996. On the identifiability and distinguishability of nonlinear parametric models. *Math. Comput. Simulation (ISSN: 0378-4754)* 42, 125–134. [http://dx.doi.org/10.1016/0378-4754\(95\)00123-9](http://dx.doi.org/10.1016/0378-4754(95)00123-9).
- Yang, Yulin, Huang, Guoquan, 2019. Observability analysis of aided INS with heterogeneous features of points, lines, and planes. *IEEE Trans. Robot.* 35 (6), 1399–1418.